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Nonlinear Buckling of a Cylindrical Shell

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ABSTRACT

Nonlinear Buckling and Post-Buckling Analysis of a Thin-Walled Cylindrical Shell Using CalculiX

This study investigates the nonlinear buckling and post-buckling behavior of a thin-walled cylindrical shell using the open-source finite element software CalculiX. Thin-walled cylindrical shells are widely used in engineering applications due to their high load-carrying capacity and lightweight structural form. However, these structures are highly sensitive to buckling instability under compressive loading, which makes the accurate prediction of critical load and post-buckling response an important design requirement.

In this project, the cylindrical shell geometry was defined with a radius of 30 mm, a wall thickness of 1 mm, and a length of 80 mm. The material was assumed to be steel with a Young's modulus of 210000 MPa and a Poisson's ratio of 0.3. First, analytical calculations were performed to determine the cross-sectional area, axial stress, critical buckling stress, and corresponding critical buckling load. Based on the target critical stress of approximately 4000 MPa, the analytical critical load was calculated as approximately 741 kN.

After the theoretical calculations, finite element analyses were carried out using CalculiX and CalculiX GraphiX. Initially, force-controlled nonlinear static analyses were performed. However, as the model approached the instability region, convergence problems occurred before reaching the analytical critical load. The force-controlled solution converged up to approximately 671 kN, indicating that the structure entered an unstable buckling region where the equilibrium path could not be followed only by increasing the applied force.

To overcome this limitation, a displacement-controlled analysis was implemented. In this method, prescribed displacement was applied to the loading region, and the corresponding reaction forces were extracted from the CalculiX DAT output file. The load–displacement curve was then obtained using an XY scatter plot, where the displacement values were

placed on the horizontal axis and the reaction force values on the vertical axis. The results showed that displacement-controlled analysis is more suitable for tracing the nonlinear response near the buckling region and evaluating the post-buckling behavior of thin-walled cylindrical shells.

This study demonstrates the importance of selecting an appropriate loading strategy in nonlinear buckling analysis. It also shows that analytical critical load calculations, finite element simulations, convergence behavior, and load–displacement curves must be evaluated together to understand the instability response of thin-walled cylindrical shell structures.

ÖZET

CalculiX Kullanılarak İnce Cidarlı Silindirik Kabuğun Doğrusal Olmayan Burkulma ve Burkulma Sonrası Analizi

Bu çalışmada, ince cidarlı silindirik bir kabuğun doğrusal olmayan burkulma ve burkulma sonrası davranışı sayısal olarak incelenmiştir. Silindirik kabuk yapılar; düşük ağırlıklarına rağmen yüksek taşıma kapasitesine sahip olmaları nedeniyle makine, havacılık, enerji ve inşaat mühendisliği uygulamalarında yaygın olarak kullanılmaktadır. Ancak bu tür ince cidarlı yapılar, basma yükleri altında burkulma kararsızlığına karşı oldukça hassastır. Bu nedenle, silindirik kabukların kritik yük seviyelerinin ve burkulma sonrası davranışlarının doğru şekilde belirlenmesi mühendislik tasarımı açısından önemlidir.

Çalışmada öncelikle silindirik kabuğun temel geometrik özellikleri tanımlanmıştır. Model; yarıçap, cidar kalınlığı, uzunluk, elastisite modülü ve Poisson oranı gibi temel parametreler kullanılarak oluşturulmuştur. Kesit alanı ve eksenel gerilme değerleri analitik olarak hesaplanmış, kritik burkulma gerilmesi ve buna karşılık gelen kritik yük teorik formüller yardımıyla belirlenmiştir. Yapılan hesaplamalarda kritik burkulma gerilmesi yaklaşık 4000 MPa mertebesinde kabul edilmiş ve buna karşılık gelen kritik yük yaklaşık 741 kN olarak elde edilmiştir.

Sayısal analizler açık kaynaklı sonlu elemanlar yazılımı CalculiX ve ön/son işlemci olarak kullanılan CalculiX GraphiX aracılığıyla gerçekleştirilmiştir. İlk aşamada modelin doğruluğunu kontrol etmek amacıyla kuvvet kontrollü analizler yapılmış, yükleme ve sınır şartlarının modele doğru şekilde aktarıldığı doğrulanmıştır. Daha sonra silindirik kabuk modeli üzerinde remote point ve coupling yaklaşımı kullanılarak yük uygulaması gerçekleştirilmiştir. Kuvvet kontrollü doğrusal olmayan analizlerde modelin kritik burkulma bölgesine yaklaştığında yakınsama problemi yaşadığı görülmüştür. Bu durum, burkulma problemlerinde yük kontrollü çözümün kararsız denge yollarını takip etmekte yetersiz kalabileceğini göstermiştir.

Bu nedenle çalışmanın devamında deplasman kontrollü analiz yaklaşımı kullanılmıştır. Bu yöntemde yapıya doğrudan belirli bir yer değiştirme uygulanmış ve oluşan reaksiyon

kuvvetleri CalculiX çıktı dosyalarından elde edilmiştir. Elde edilen reaksiyon kuvveti ve yer değiştirme değerleri kullanılarak yük–deplasman eğrisi oluşturulmuştur. Grafik oluşturulurken yer değiştirme değerleri yatay eksen, reaksiyon kuvvetleri ise düşey eksen kullanılmış ve sonuçlar XY scatter grafik yöntemiyle değerlendirilmiştir. Böylece silindirik kabuğun yük altındaki rijitlik değişimi, kritik bölgeye yaklaşması ve burkulma sonrası davranış eğilimi incelenmiştir.

Sonuç olarak, kuvvet kontrollü analizlerin kritik burkulma bölgesine kadar sınırlı bilgi verdiği, ancak burkulma sonrası davranışı izlemek için deplasman kontrollü yöntemin daha uygun olduğu görülmüştür. Elde edilen sonuçlar, ince cidarlı silindirik kabukların doğrusal olmayan davranışlarının sayısal olarak incelenmesinde yükleme yöntemi, yakınsama davranışı ve reaksiyon kuvvetlerinin doğru yorumlanmasının büyük önem taşıdığını göstermektedir.

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LIST OF ABBREVIATIONS

Abbreviations

Abbreviation	Definition
FEA	Finite Element Analysis
FEM	Finite Element Method
CGX	CalculiX GraphiX; pre-processing and post-processing tool
CCX	CalculiX CrunchiX; finite element solver
LBA	Linear Bifurcation Analysis
GNIA	Geometrically Nonlinear Analysis with Imperfections
KDF	Knockdown Factor
DOF	Degree of Freedom
FRD	CalculiX result file format used for post-processing
INP	CalculiX input file format
MSH	Mesh file containing node and element information
NAM	Node set file used for boundary and loading regions

Symbols Used in the Analytical Formulations

Symbol	Definition	Unit
E	Young's modulus	MPa
ν	Poisson's ratio	-
ρ	Density	kg/mm ³ or tonne/mm ³ depending on model unit system
R	Cylinder radius	mm
L	Cylinder length	mm

t	Shell wall thickness	mm
R/t	Radius-to-thickness ratio	-
L/R	Length-to-radius ratio	-
σ_{cr}	Classical critical buckling stress	MPa
P_{cr}	Classical critical axial buckling load	N
$P_{imperfect}$	Buckling load of the imperfect cylindrical shell	N
$P_{perfect}$	Buckling load of the perfect cylindrical shell	N
P_{real}	Estimated real buckling load after applying the knockdown factor	N
λ	Load factor / eigenvalue multiplier	-
u	Displacement vector	mm
Δu	Incremental displacement vector	mm
$\Delta \lambda$	Incremental load factor	-
$R(u, \lambda)$	Residual force vector in nonlinear equilibrium equation	N
F_{int}	Internal force vector	N
F_{ext}	External force vector	N
K_T	Tangent stiffness matrix	N/mm
$\det(K_T)$	Determinant of the tangent stiffness matrix	-
N_{fix}	Node set used for the fixed boundary condition	-
N_{load}	Node set used for the load application region	-
X, Y, Z	Global coordinate directions used for loading cases	-

1. INTRODUCTION

1.1. Definition of Buckling

Buckling is a structural instability phenomenon that occurs when a structural member subjected to compressive loading suddenly loses its stable equilibrium configuration. In this condition, the structure may experience a rapid increase in lateral displacement even though the applied load does not increase significantly. Therefore, buckling is not only related to material failure, but also to the loss of stability of the structural system.

In conventional strength problems, failure is generally associated with the stress level reaching the yield strength or ultimate strength of the material. However, in buckling problems, a structure can become unstable before the material reaches its strength limit. This behavior is especially important for slender columns, plates, and thin-walled shells, where the geometry of the structure plays a major role in the load-carrying capacity.

For a cylindrical shell, buckling usually occurs under axial compression, external pressure, bending, torsion, or combined loading conditions. Due to their thin-walled geometry, cylindrical shells can carry membrane loads efficiently, but they are highly sensitive to instability. A small change in geometry, loading condition, boundary condition, or imperfection can significantly affect the critical buckling load and the resulting buckling mode shape.

The critical buckling load is the load level at which the structure begins to lose stability. Before this point, the structure follows a stable equilibrium path. After reaching the critical point, the structure may move suddenly to a different deformation pattern. This behavior is commonly observed as a rapid change in displacement or a snap-through response. For this reason, buckling analysis requires not only stress evaluation but also the investigation of equilibrium, stiffness, and deformation behavior.

In this study, buckling is considered as the instability response of a thin-walled cylindrical shell under compressive loading. The main focus is to determine the critical load region and

to examine the nonlinear load–displacement behavior of the shell using finite element analysis.



Figure 1: Real Life Buckling Example

1.2. Buckling Behavior in Engineering Structures

Buckling behavior is one of the most important stability problems encountered in engineering structures. Many structural components are designed to carry compressive loads while maintaining their original shape and stiffness. However, when the applied compressive load reaches a critical level, the structure may lose its stable configuration and experience a sudden change in deformation pattern. This instability can occur even when the stress level is lower than the material yield strength, which makes buckling different from conventional strength-based failure.

In engineering applications, buckling is commonly observed in slender columns, thin plates, cylindrical shells, pressure vessels, pipelines, aerospace structures, and lightweight mechanical components. These structures are often designed with reduced thickness in order to decrease weight and material usage. Although this design approach provides efficiency, it also increases sensitivity to instability. As a result, thin-walled structures may fail due to buckling before reaching their full material strength capacity.

The buckling behavior of a structure depends on several factors, including geometry, thickness, material properties, boundary conditions, loading type, and initial imperfections. For example, a small geometric imperfection or a slight eccentricity in the applied load can significantly reduce the critical buckling load of a thin-walled shell. Therefore, ideal analytical solutions may provide only an approximate prediction, while the actual structural response can be more complex.

In practical engineering design, understanding the load–displacement behavior of a structure is essential for evaluating buckling. Before buckling occurs, the structure usually shows a relatively stable and predictable stiffness response. Near the critical load, the stiffness may decrease and the structure may enter an unstable equilibrium region. After this point, large displacements may occur with a small increase in load, or the structure may follow a different post-buckling path. This behavior is especially important for shell-type structures, where snap-through or sudden deformation can occur.

For this reason, buckling analysis is not only used to estimate the critical load, but also to understand the nonlinear response of the structure near and after instability. In this study, the buckling behavior of a thin-walled cylindrical shell is investigated using finite element

analysis. The response of the model under compressive loading is evaluated through force-controlled and displacement-controlled analysis methods. The obtained load–displacement curve is used to interpret the stiffness change, instability region, and post-buckling tendency of the cylindrical shell.



Figure 2: NASA engineers prepare a test article for a series of tests at NASA's Marshall Space Flight Center.

1.3. Historical Development of Buckling Theory

The study of buckling has an important place in the history of structural mechanics. The first major theoretical contribution to buckling theory was made by Leonhard Euler in the eighteenth century. Euler developed the classical column buckling formulation, which

describes the critical load at which a slender column loses its stable straight configuration under axial compression. This theory formed the foundation of stability analysis and showed that structural failure may occur due to instability rather than material yielding.

After Euler's work, buckling theory was expanded to include more complex structural members such as beams, plates, and shells. As engineering structures became lighter and more efficient, the stability of thin-walled components became increasingly important. Classical theories were developed to estimate the critical buckling loads of plates and cylindrical shells under different loading and boundary conditions. These analytical solutions provided important insight into the relationship between geometry, material properties, stiffness, and critical load.

However, the buckling behavior of shell structures was found to be more complicated than simple column buckling. Thin-walled cylindrical shells are highly sensitive to initial geometric imperfections, loading eccentricities, and boundary conditions. Experimental studies showed that the actual buckling loads of shell structures could be significantly lower than the values predicted by ideal analytical formulas. This difference between theoretical and experimental results made shell buckling one of the most challenging topics in structural stability.

With the development of computational methods, finite element analysis became an important tool for investigating buckling problems. Numerical models allowed engineers to analyze complex geometries, realistic boundary conditions, material properties, and nonlinear deformation behavior. Linear buckling analysis provided an estimation of ideal critical loads, while nonlinear analysis made it possible to examine large displacement effects, instability regions, and post-buckling response.

In modern engineering analysis, buckling is therefore studied using both analytical and numerical methods. Analytical formulas are useful for obtaining initial estimates and understanding the basic mechanics of instability. However, nonlinear finite element analysis is required when the structure experiences large deformations, convergence difficulties, or snap-through behavior. In this project, the historical development of buckling theory is reflected in the analysis approach: classical buckling calculations are first used to estimate the critical load, and then nonlinear finite element simulations are performed using CalculiX to investigate the load-displacement response of the thin-walled cylindrical shell.

1.4. Linear and Nonlinear Buckling Analysis

Buckling analyses are generally examined using two main approaches: linear buckling analysis and nonlinear buckling analysis. Linear buckling analysis assumes that the structure experiences small deformations before buckling and that the material behaves in a linear elastic manner. In the finite element method, this analysis is usually solved as an eigenvalue problem. As a result, the critical load factor and the corresponding buckling mode shapes are obtained. This method is useful for quickly estimating the first buckling load of an ideal structure.

However, linear buckling analysis is based on ideal geometry, perfect boundary conditions, and small deformation assumptions. In real structures, geometric imperfections, loading eccentricity, boundary condition variations, and manufacturing defects can significantly affect the critical buckling load. Thin-walled cylindrical shells are especially sensitive to these imperfections. Therefore, linear buckling analysis generally provides an ideal critical load and may not fully represent the actual structural behavior.

Nonlinear buckling analysis considers the behavior of the structure under large deformations and accounts for stiffness changes during loading. In this type of analysis, geometric nonlinearity is included. As the structure deforms, its equilibrium state and stiffness change. Therefore, nonlinear analysis allows a more realistic investigation of instability near the critical load region and post-buckling behavior.

One of the main difficulties in buckling problems is that the structure may not follow a smooth and continuous equilibrium path after reaching the critical point. Instead, it may suddenly jump from one equilibrium branch to another. This behavior can be observed in the load–displacement curve as snap-through or sudden instability. For this reason, buckling behavior should not be evaluated only by stress values, but also by the load–displacement relationship and stability response.

In force-controlled analyses, the applied load is increased step by step. This approach is generally sufficient in the stable pre-buckling region. However, near the critical region, the structure may experience increasing displacement while the required reaction force decreases or becomes nearly constant. In such cases, a solution method based only on increasing the applied load may not be able to follow the unstable equilibrium path, leading to convergence problems. In this study, the force-controlled analysis approached the critical region, but the solution did not converge beyond a certain load level.

For this reason, a displacement-controlled analysis approach was used to investigate the post-buckling behavior. In this method, a prescribed displacement was applied to the structure, and the corresponding reaction forces were obtained from the CalculiX output files. In this way, the load–displacement curve was constructed more accurately, and the nonlinear buckling behavior of the cylindrical shell was evaluated.

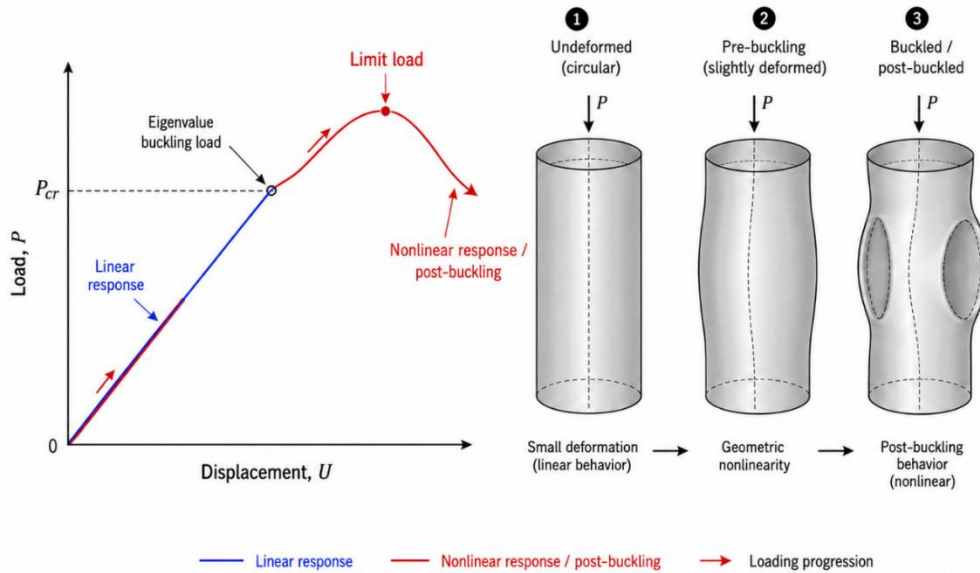


Figure 3: Comparison of linear and nonlinear buckling behavior through the load–displacement response and deformation stages of a cylindrical shell.

1.5. Finite Element Analysis in Buckling Problems

Finite Element Analysis is one of the most widely used numerical methods for investigating buckling problems in engineering structures. Buckling behavior is strongly affected by geometry, boundary conditions, loading type, mesh quality, and nonlinear deformation effects. For this reason, analytical formulas are often useful for preliminary calculations, but they may not be sufficient to represent the actual behavior of complex structures, especially thin-walled shells.

In buckling problems, finite element analysis allows the structure to be divided into smaller elements and solved numerically under defined loading and boundary conditions. This approach makes it possible to study different structural responses such as stress distribution, displacement field, deformation shape, reaction forces, and critical instability behavior. For thin-walled cylindrical shells, finite element analysis is especially useful because these structures are sensitive to imperfections and may show nonlinear behavior before reaching the ideal analytical critical load.

Finite element buckling analysis can be performed using different solution strategies. Linear buckling analysis estimates the ideal critical load and buckling mode shape based on the initial stiffness of the structure. Nonlinear buckling analysis, on the other hand, considers the change in stiffness due to deformation and provides a more realistic representation of the structural response. In nonlinear analysis, geometric nonlinearity is important because the deformation of the structure directly affects its load-carrying capacity.

In this project, finite element analysis was performed using CalculiX and CalculiX GraphiX. CalculiX was used as the solver, while CalculiX GraphiX was used for geometry preparation, mesh generation, and visualization of results. The cylindrical shell model was analyzed under compressive loading conditions, and both force-controlled and displacement-controlled solution approaches were examined. The force-controlled analysis provided information about the response of the model up to the instability region. However, convergence difficulties occurred near the critical buckling region.

To investigate the nonlinear response more effectively, displacement-controlled analysis was used. In this method, a prescribed displacement was applied to the loading region, and the

corresponding reaction forces were extracted from the output files. These reaction force values were then used to construct the load–displacement curve. This curve provided an important basis for evaluating the stiffness change, instability behavior, and post-buckling tendency of the thin-walled cylindrical shell.

Therefore, finite element analysis plays a central role in this study. It connects the theoretical critical buckling calculations with the actual numerical response of the model. By using CalculiX, the buckling behavior of the cylindrical shell could be investigated in a more detailed and practical way than using analytical formulas alone.

In addition to the analytical calculations and nonlinear static analyses, a linear eigenvalue buckling analysis was also considered as an intermediate numerical reference in this study. Eigenvalue buckling analysis provides an ideal critical load factor and the corresponding buckling mode shape of the structure under a defined reference load. Although this approach does not include geometric imperfections or large deformation effects, it is useful for estimating the theoretical buckling tendency of the model and for identifying the expected instability mode. Therefore, the eigenvalue result was used together with the analytical critical load and the nonlinear load–displacement response in order to compare the ideal buckling prediction with the more realistic nonlinear behavior obtained from CalculiX.

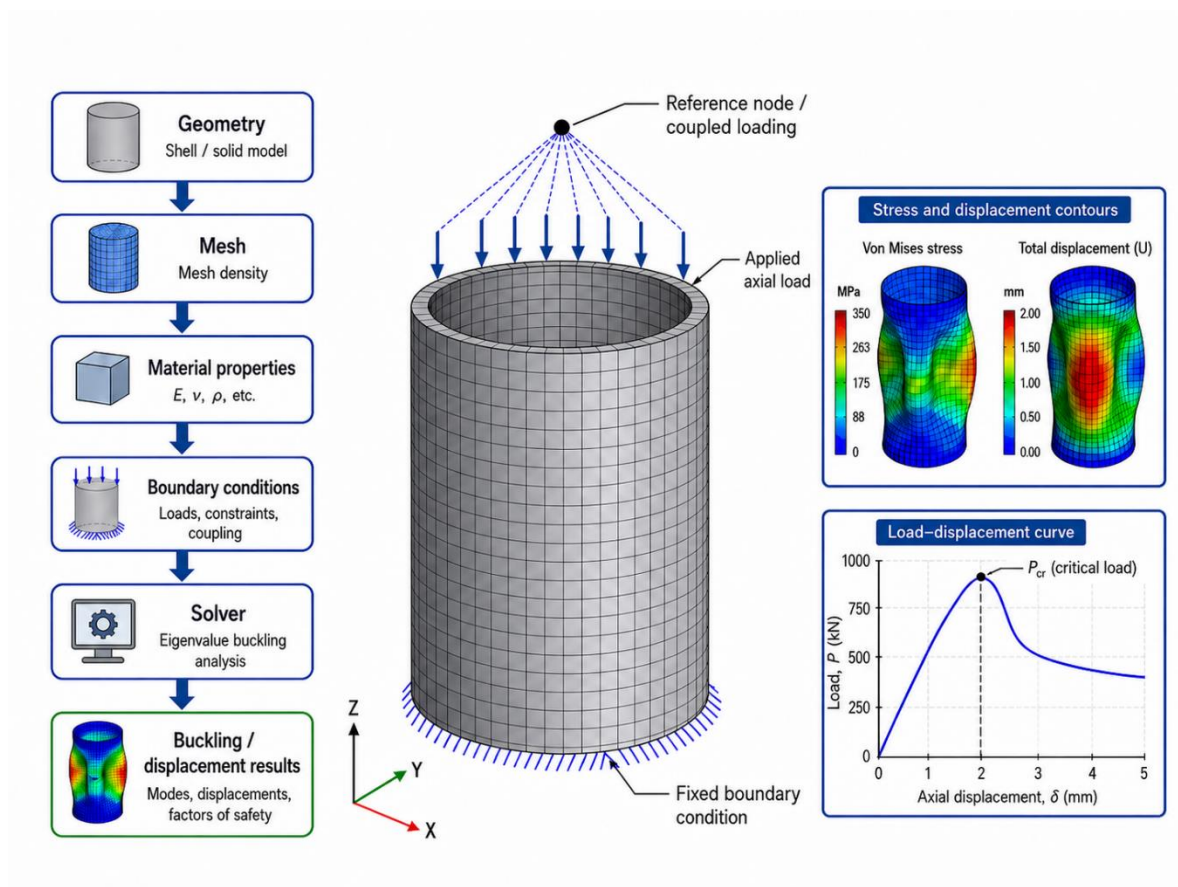


Figure 4: General finite element analysis workflow for buckling problems, including geometry, meshing, boundary conditions, loading, and result evaluation.

2. LITERATURE REVIEW

Buckling has been one of the fundamental topics in structural mechanics because it represents a stability failure rather than a conventional material failure. Early studies on buckling mainly focused on slender columns under axial compression. Euler's classical column buckling theory formed the basis of structural stability analysis and showed that a member may lose its stable configuration before reaching the material yield strength. This idea later became the foundation for the analysis of more complex structural elements such as beams, plates, and shells.

As engineering structures became lighter and more efficient, the buckling behavior of thin-walled members gained significant importance. Thin-walled cylindrical shells are widely used in aerospace structures, pipelines, pressure vessels, storage tanks, and mechanical components due to their high load-carrying capacity with relatively low weight. However, these structures are also highly sensitive to instability under compressive loading. A small geometric imperfection, eccentric loading condition, or change in boundary conditions may significantly reduce the actual buckling load compared with the ideal analytical prediction.

Classical shell buckling theories were developed to estimate the critical load of cylindrical shells under different loading conditions. These theories provide useful analytical expressions for the critical buckling stress and critical load. In the case of thin-walled cylindrical shells under axial compression, the critical buckling stress is commonly related to the elastic modulus, Poisson's ratio, shell thickness, and shell radius. Such analytical formulas are useful for obtaining an initial design estimate and for understanding the main parameters affecting buckling behavior. However, ideal analytical solutions generally assume perfect geometry, uniform loading, and simplified boundary conditions.

Experimental and numerical studies have shown that cylindrical shell buckling is strongly affected by imperfections. In a geometrically perfect shell, the structure may follow a higher ideal load path before reaching the critical instability point. However, in a geometrically imperfect shell, the load-carrying capacity may decrease significantly, and the nonlinear response may start earlier. This difference between perfect and imperfect shell behavior is one of the main reasons why shell buckling is considered a challenging topic in structural stability.

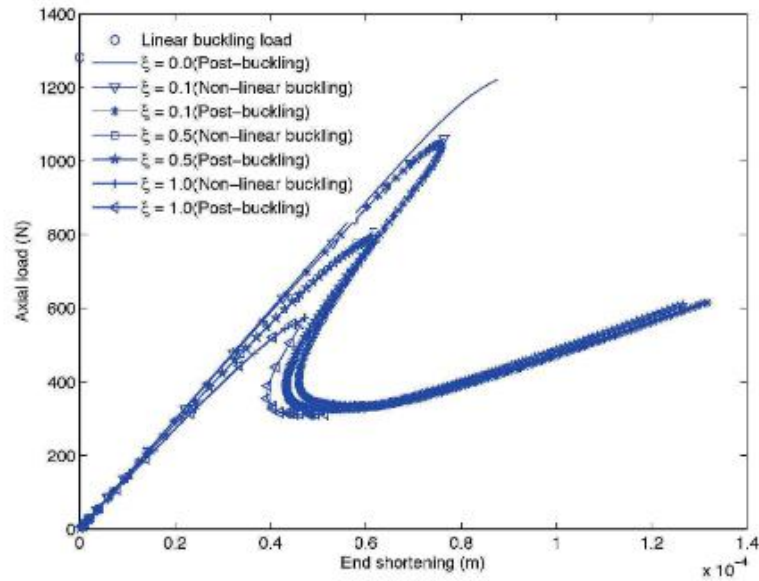


Figure 5: Load–deflection behavior of geometrically perfect and imperfect cylindrical shells under compressive loading

The load–deflection response shown in Figure 2.1 illustrates the importance of imperfection sensitivity in cylindrical shell buckling. While the geometrically perfect shell can reach a higher theoretical critical load, the imperfect shell shows a lower load-carrying capacity and a more gradual nonlinear response. This behavior is important for practical engineering design because real shell structures almost always contain some level of imperfection due to manufacturing, welding, handling, or boundary condition effects.

Experimental investigations have played an important role in understanding shell buckling behavior. Physical tests on cylindrical shells have shown that the actual deformation pattern may differ from ideal theoretical assumptions. During compression tests, local inward and outward deformations may develop on the shell surface, and the final buckling shape can be affected by imperfections, loading uniformity, boundary support, and shell thickness. These experimental observations support the need for nonlinear numerical analysis in addition to analytical calculations.

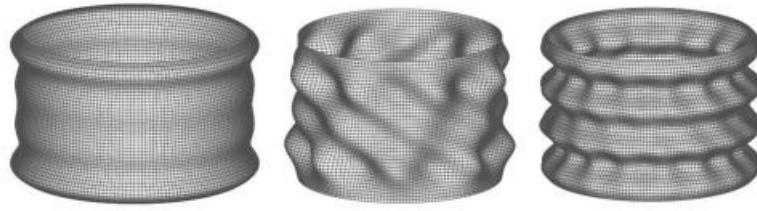


Figure 6: Numerical buckling deformation patterns of cylindrical shells obtained from finite element analysis.

With the development of computational methods, finite element analysis became an important tool for investigating buckling problems. Linear buckling analysis is generally used to estimate ideal critical loads and buckling mode shapes. In this method, the analysis is based on the initial stiffness of the structure and provides a buckling factor corresponding to the applied reference load. Although linear buckling analysis is computationally efficient, it cannot fully represent large deformation effects, nonlinear stiffness changes, convergence difficulties, or post-buckling response.

Finite element simulations are also useful for visualizing buckling mode shapes and deformation patterns of cylindrical shells. Different buckling modes may occur depending on geometry, mesh density, boundary conditions, loading type, and imperfection pattern. These numerical deformation shapes help engineers understand where instability starts and how the shell loses its load-carrying capacity.

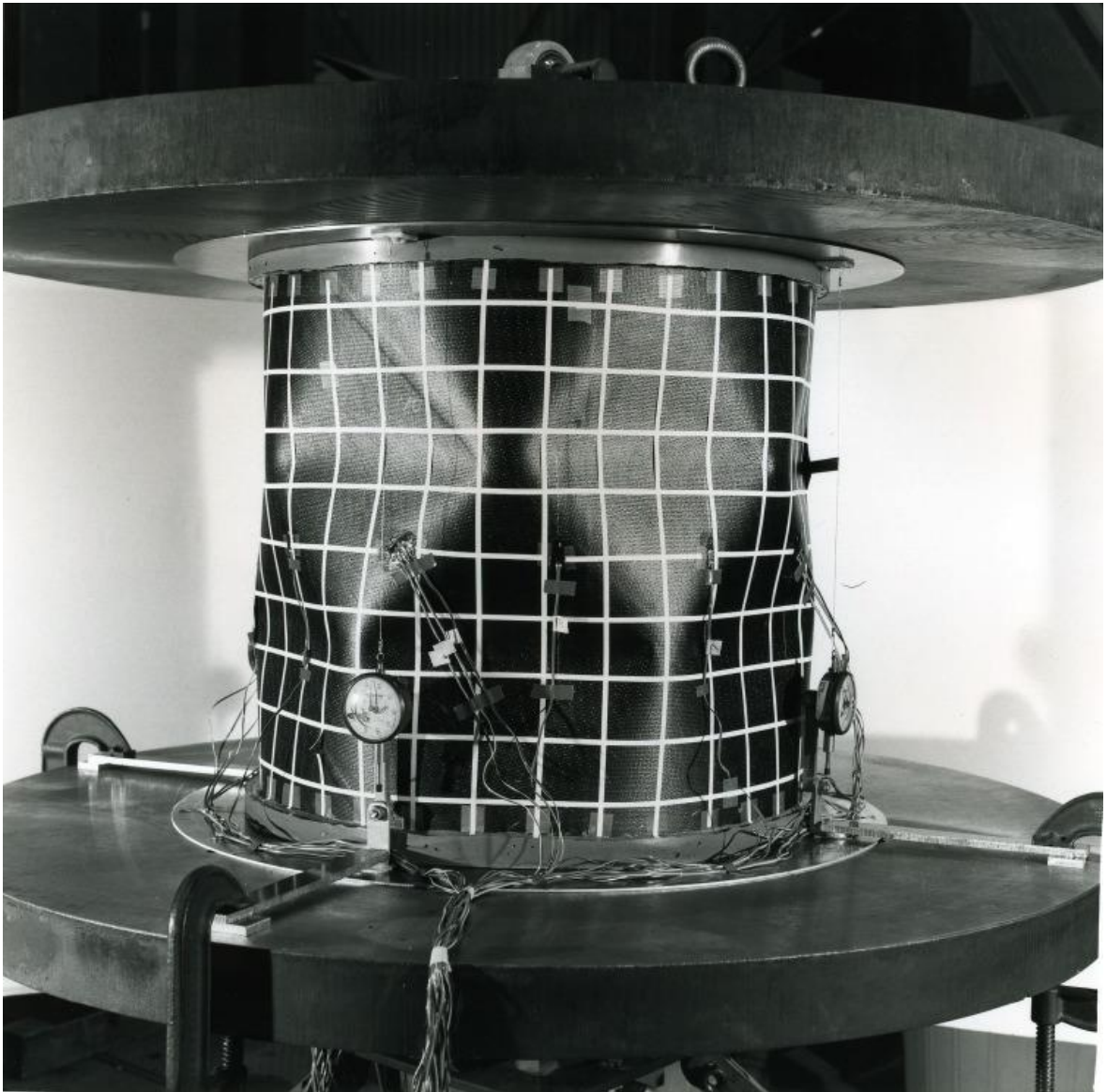


Figure 7: Experimental investigation of cylindrical shell buckling under axial compression.

Nonlinear buckling analysis provides a more realistic approach by considering geometric nonlinearity and large deformation behavior. In nonlinear analysis, the load–displacement response of the structure can be followed as the deformation increases. This makes it possible to observe stiffness reduction, instability regions, snap-through behavior, and post-buckling tendencies. However, nonlinear buckling analysis is also more difficult because the numerical solution may lose convergence near the critical instability region.

In previous studies, both force-controlled and displacement-controlled methods have been used to investigate nonlinear buckling behavior. In force-controlled analysis, the applied load is gradually increased and the corresponding displacement response is calculated. This method is useful before the critical region, but it may fail to follow the equilibrium path when the structure enters an unstable response region. In displacement-controlled analysis, displacement is prescribed and the reaction force is obtained from the numerical model. This approach is often more suitable for constructing the load–displacement curve near the buckling region.

In this thesis, the literature background is applied to the nonlinear buckling analysis of a thin-walled cylindrical shell using CalculiX. The analytical critical load calculation is used as a reference, while finite element simulations are performed to investigate the actual nonlinear response of the model. The force-controlled analysis showed convergence limitations near the instability region. Therefore, displacement-controlled analysis was used to obtain reaction force data and construct the load–displacement curve. This approach allows the buckling behavior of the cylindrical shell to be evaluated in a more complete and practical manner.

3. ANALYTICAL FORMULATIONS

3.1. 3.1 Cylindrical Shell Geometry and Basic Assumptions

The analytical formulation of the buckling problem begins with the definition of the cylindrical shell geometry. In this study, the investigated structure is modeled as a thin-walled cylindrical shell with a circular cross-section. The main geometrical parameters of the shell are the radius (R), wall thickness (t), and length (L). These parameters directly affect the stiffness, stress distribution, and critical buckling load of the structure.

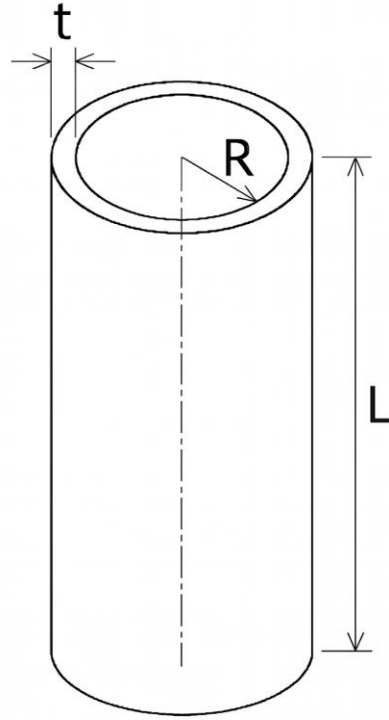


Figure 8: Geometrical parameters of the thin-walled cylindrical shell model.

For the analytical calculations, the cylindrical shell is assumed to have a uniform wall thickness along its length. The cross-section is considered circular, and the shell material is assumed to be homogeneous, isotropic, and linearly elastic. In this project, the material

is steel, with a Young's modulus of ($E = 210000$) MPa and a Poisson's ratio of ($\nu = 0.3$). These assumptions are commonly used in classical elastic buckling formulations. The geometrical dimensions used in the calculations are given as ($R = 30$) mm, ($t = 1$) mm, and ($L = 80$) mm. Since the wall thickness is much smaller than the radius, the structure can be treated as a thin-walled cylindrical shell. The thickness-to-radius ratio is

$$\frac{t}{R} = \frac{1}{30} = 0.0333 \quad (7)$$

which indicates that the shell has a relatively slender wall compared with its radius. This thin-walled characteristic makes the structure sensitive to buckling under compressive loading.

In the analytical formulation, the shell is assumed to be geometrically perfect. This means that no initial geometric imperfections, local dents, ovalization, or manufacturing defects are considered in the theoretical calculations. The loading is assumed to be applied in the axial direction, and the stress distribution is assumed to be uniform over the cross-section before buckling occurs. These assumptions simplify the analytical solution and allow the critical buckling load to be estimated using classical shell buckling equations.

However, it should be noted that real cylindrical shells may contain imperfections and nonuniformities. Therefore, the analytical formulation provides an ideal reference value rather than an exact representation of the actual nonlinear response. In this thesis, the analytical calculations are used as a reference for comparison with the finite element analysis results obtained from CalculiX. The numerical model is then used to investigate the nonlinear load–displacement behavior and convergence response of the cylindrical shell near the buckling region.

3.2. Analysis of Formulations

After defining the cylindrical shell geometry and the basic assumptions, the analytical formulations used in this study are presented in this section. These formulations are used to estimate the stress level and the theoretical critical buckling load of the thin-walled

cylindrical shell. The analytical results provide a reference for evaluating the finite element analysis results obtained from CalculiX.

The first parameter required for the stress calculation is the cross-sectional area of the cylindrical shell. Since the structure is modeled as a hollow circular section, the cross-sectional area is calculated by subtracting the inner circular area from the outer circular area. The area is expressed as

$$\begin{aligned}
 A &= \pi(R^2 - R_i^2) \\
 R_i &= R - t \\
 A &= \pi(R^2 - (R - t)^2)
 \end{aligned}
 \tag{1}$$

where (R) is the outer radius and (t) is the wall thickness. In this study, the radius and thickness are defined as (R = 30) mm and (t = 1) mm. Therefore, the cross-sectional area becomes

$$A = \pi [30^2 - (30 - 1)^2]$$

$$A = \pi [30^2 - 29^2]$$

$$A = 185.35 \text{ mm}^2$$

The average axial stress generated by an axial compressive load can be calculated using:

$$\sigma = \frac{P}{A}
 \tag{1}$$

where:

σ =average axial stress

P =applied axial load

A =cross-sectional area

This equation is used to relate the applied load to the average compressive stress in the shell before buckling occurs. Since the material is assumed to remain elastic in the analytical formulation, the stress is evaluated within the elastic range.

For a thin-walled cylindrical shell under axial compression, the classical elastic critical buckling stress is estimated using:

$$\sigma_{cr} = \frac{E}{\sqrt{3(1 - \nu^2)}} \left(\frac{t}{R} \right) \quad (5)$$

where (E) is Young's modulus, ν is Poisson's ratio, (t) is the wall thickness, and (R) is the radius of the cylindrical shell. This equation shows that the critical buckling stress is directly proportional to the elastic modulus and the thickness-to-radius ratio. Therefore, thinner shells with smaller (t/R) ratios are more sensitive to buckling.

In this project, the material is steel with

$$E = 210000 \text{ MPa}$$

$$\nu = 0.3$$

Using these values, the classical formulation gives a critical buckling stress close to the reference value used in the project. For the analytical comparison, the critical buckling stress is taken as approximately

$$\sigma_{cr} \approx 4236.6 \text{ MPa}$$

The corresponding analytical critical buckling load is calculated by multiplying the critical buckling stress by the cross-sectional area:

$$P_{cr} = A \sigma_{cr} \quad (3)$$

Substituting the values,

$$P_{cr} = 4000 * 185.35$$

$$P_{cr} \approx 741400 \text{ N} \approx 741.4 \text{ kN}$$

This value represents the analytical reference critical load for the cylindrical shell. It is important to note that this value is based on ideal assumptions such as perfect geometry, uniform loading, elastic material behavior, and simplified boundary conditions. Therefore, it should not be interpreted as the exact buckling load of the numerical model. Instead, it is used as a theoretical reference for comparison with the nonlinear finite element analysis results.

In the finite element analysis, the force-controlled nonlinear solution reached approximately 671 kN before convergence difficulties occurred near the instability region. This value is lower than the analytical reference load of approximately 741.4 kN. The difference is expected because nonlinear numerical analysis is affected by boundary conditions, loading method, mesh behavior, geometric stiffness changes, and convergence limitations near buckling.

The analytical formulations used in this section therefore provide the theoretical basis of the study. They define the relationship between geometry, material properties, stress, and critical buckling load. In the following numerical analysis sections, these analytical results are used to evaluate the nonlinear response and load–displacement behavior obtained from CalculiX.

3.3. Linear Eigenvalue Buckling Formulation

In addition to the classical analytical buckling calculation, a linear eigenvalue buckling formulation was used as a numerical reference for estimating the ideal buckling load of the cylindrical shell model. Linear eigenvalue buckling analysis is based on the assumption of small deformation, linear elastic material behavior, and an initially perfect geometry. Therefore, it provides an ideal critical load prediction rather than the actual nonlinear buckling response of the structure.

In finite element buckling analysis, the linear buckling problem can be expressed as an eigenvalue problem:

$$(\mathbf{K} + \lambda \mathbf{K}_G) \boldsymbol{\phi} = \mathbf{0} \quad (4)$$

where $\mathbf{K}$ is the linear elastic stiffness matrix, $\mathbf{K}_G$ is the geometric stiffness matrix generated by the applied reference load, λ is the eigenvalue or load multiplier, and $\boldsymbol{\phi}$ is the corresponding buckling mode shape. The eigenvalue λ shows how many times the applied reference load must be multiplied to reach the ideal critical buckling load.

The critical buckling load obtained from the eigenvalue analysis is calculated as:

$$P_{cr,eigen} = \lambda P_{ref} \quad (4)$$

where P_{ref} is the reference load applied in the eigenvalue buckling model.

In this project, a reference load of:

$$P_{ref} = 10 \text{ kN}$$

was used in the linear eigenvalue buckling analysis. The first eigenvalue obtained from CalculiX was approximately:

$$\lambda_1 = 81.67$$

Therefore, the ideal eigenvalue buckling load was calculated as:

$$P_{cr,eigen} = 81.67 \times 10$$

$$P_{cr,eigen} = 816.7 \text{ kN}$$

This value represents the ideal elastic buckling load predicted by the linear eigenvalue approach. Since the eigenvalue analysis does not include geometric imperfections, large deformation effects, nonlinear stiffness changes, or convergence behavior near the limit point, it is expected to give a higher critical load than the nonlinear static analysis.

In the nonlinear finite element analysis, the structure reached the limit region at approximately 671.5–671.7 kN. Therefore, the eigenvalue result of 816.7 kN was used as an ideal numerical reference, while the nonlinear load–displacement analysis was used to evaluate the more realistic instability response of the cylindrical shell.

4. METHODOLOGY

The methodology of this project was developed step by step using CalculiX and CGX in order to investigate the nonlinear load–displacement behavior of a cylindrical pipe model. The general workflow consisted of geometry preparation, meshing, boundary condition definition, loading method development, nonlinear static analysis, and comparison of displacement-controlled and load-controlled results.

4.1. Theoretical Background of Buckling

Buckling is a structural instability problem that occurs when a structure loses its stable equilibrium configuration under compressive or destabilizing loading. In buckling problems, failure may occur before the material reaches its yield strength. Therefore, the buckling response depends not only on the material properties, but also on the geometry, boundary conditions, loading type, and stiffness of the structure.

In this project, the buckling behavior of a cylindrical pipe model was investigated using nonlinear finite element analysis. Since cylindrical shell and pipe structures are sensitive to geometric nonlinearities and boundary conditions, a nonlinear static analysis was required instead of only using a simple linear stress analysis. The main response of interest was the relationship between the applied load and the resulting displacement.

The nonlinear behavior was evaluated using a load–displacement curve. At the beginning of the analysis, the structure behaves almost linearly, and the load increases with displacement. As the deformation increases, the stiffness of the structure decreases and the response becomes nonlinear. Eventually, the structure reaches a limit region where the load cannot be increased further in a stable way.

To investigate this behavior, two methods were used in this study: displacement-controlled analysis and load-controlled analysis. In the displacement-controlled method, prescribed displacement values were applied and the corresponding reaction forces were obtained. In the load-controlled method, load values were applied and the resulting displacements were measured. The purpose of using both methods was to compare the results and verify the load–displacement response of the model.

The load-controlled analysis was able to converge up to approximately 671.52 kN. Above this load level, the solver stopped with the error “increment size smaller than minimum,” indicating that the model reached a nonlinear limit point. The displacement-controlled analysis also showed convergence difficulty around the same region. Therefore, the approximate limit load of the cylindrical pipe model was identified as about 671.52 kN, with a corresponding displacement of approximately 1.447.

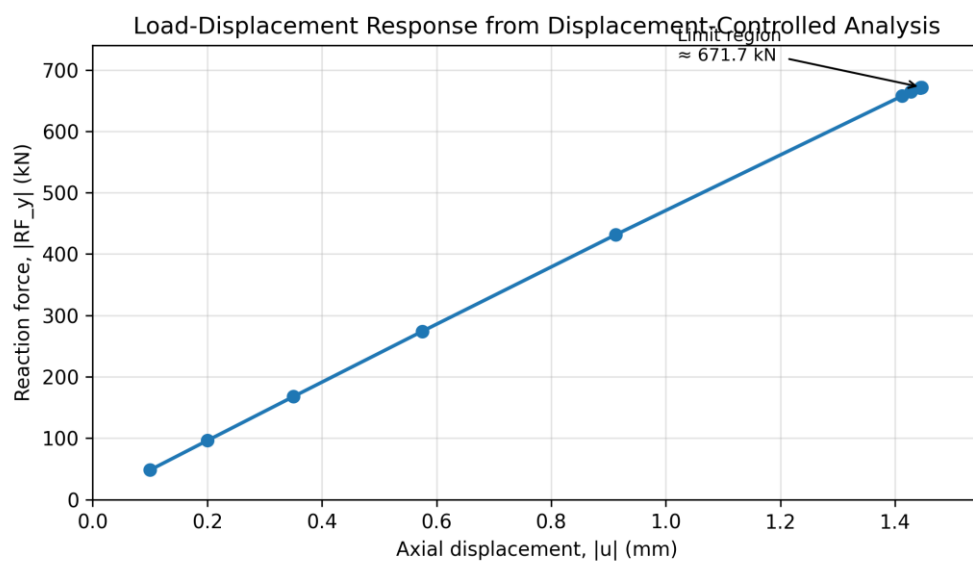


Figure 9: Load-Displacement Graph

4.2. Software Environment and General Workflow

The finite element analysis was carried out using CalculiX. CGX was used as the preprocessor and postprocessor, while CCX was used as the solver. The input files were edited manually using Notepad++, and the analyses were executed through the command line.

The general workflow used throughout the project was:

1. Geometry and mesh generation using .fbd files in CGX.
2. Creation of mesh and node/surface set files such as all.msh, fix.nam, load.nam, and load.sur.
3. Preparation of CalculiX input files .inp for different loading cases.
4. Execution of nonlinear static analyses using CCX.
5. Extraction of displacement and reaction force data from .dat and .sta files.
6. Visualization of stress, displacement, and mesh results using CGX .frd files.

The basic command sequence was:

```
Cgx-bmodel.fbd
ccxmodel-name
cgx model-name.frd
```

[Figure 1: General CalculiX/CGX workflow or command-line workflow screenshot]

4.3. Initial Shell Model Development

The analysis initially started with a shell representation of the cylindrical pipe. The shell model was useful as an initial stage because it allowed the geometry, boundary conditions, meshing procedure, and directional loading response to be tested with a relatively simple finite element model.

The original geometry was modified into a straight cylindrical shell model. CGX was used to generate the mesh and to define the required node sets. The bottom edge of the pipe was defined as the fixed region, while the opposite edge was defined as the loading region.

The main files generated for the shell model were:

- all.msh: mesh file
- fix.nam: fixed node set
- load.nam: loading node set

The steel material was defined using an elastic modulus of 210000 MPa, Poisson's ratio of 0.3, and density of 7.85E-9. Since this was a shell model, the section was defined using a shell section with a prescribed thickness.

The fixed boundary condition was applied to the fixed node set. For the shell model, both translational and rotational degrees of freedom were constrained at the fixed end.

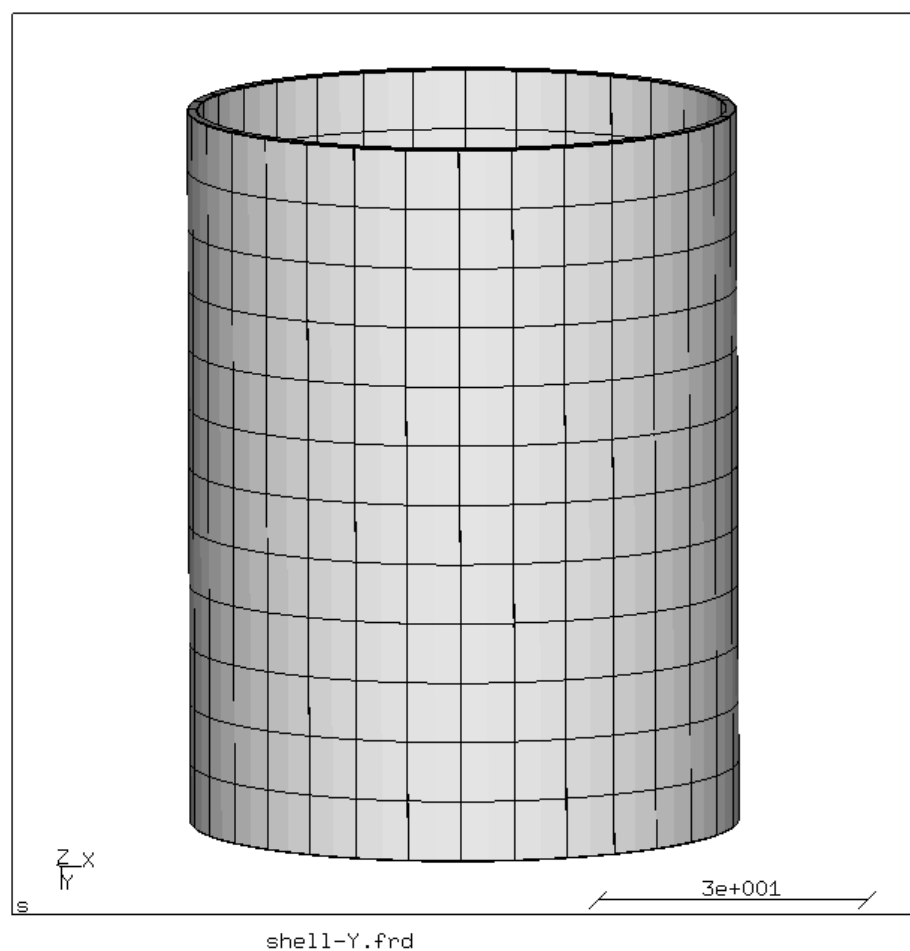


Figure 10: *Shell model geometry*

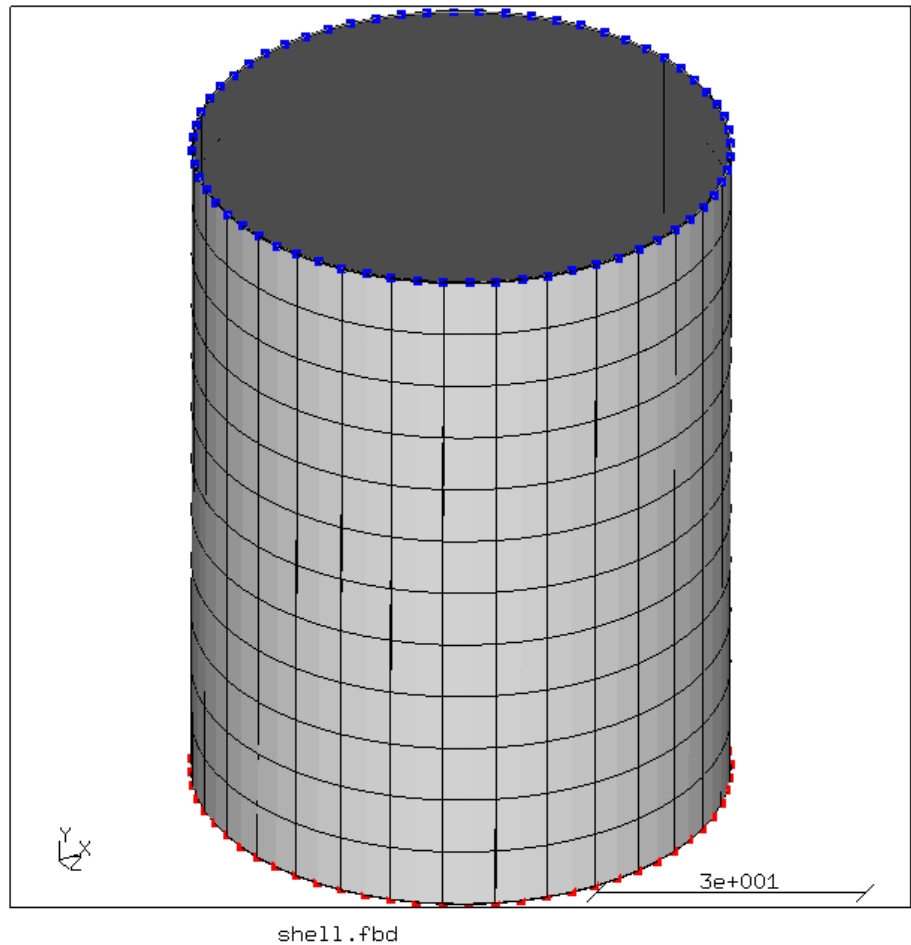


Figure 11: *Shell model mesh with fixed and loaded regions*

4.4. Shell Model Directional Loading Cases

After the shell model was prepared, three separate directional loading cases were performed in the X, Y, and Z directions. The purpose of this stage was to understand how the cylindrical shell responded under different loading directions and to compare the resulting stress distributions.

The loading directions were defined using the *CLOAD keyword. Separate input files were prepared for the X, Y, and Z loading cases.

The von Mises stress was selected as the main stress comparison parameter because it provides an equivalent stress value that is suitable for comparing the overall stress state of

the structure. The shell analysis showed that the X and Z loading directions produced higher stress levels than the Y direction. This stage helped verify the model behavior before moving to the more detailed solid model.

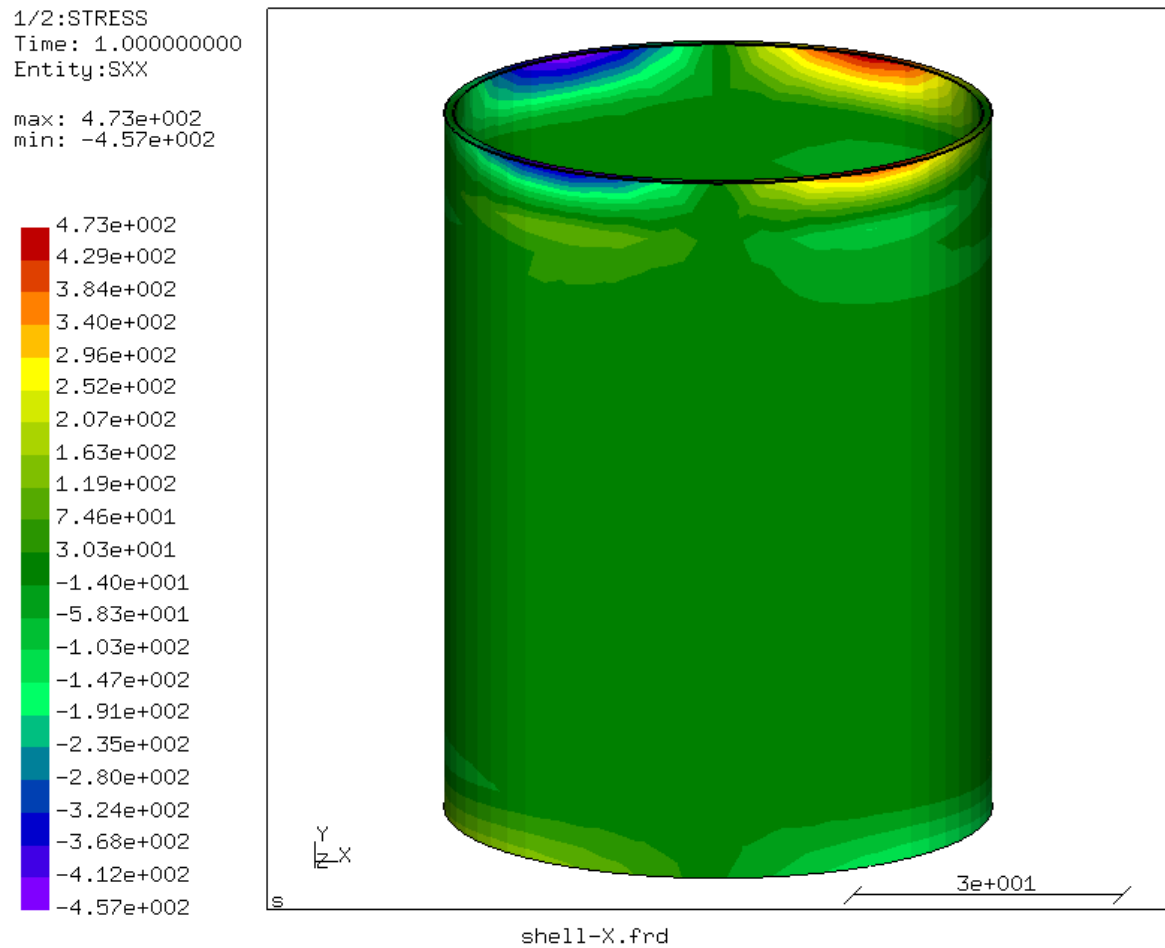


Figure 12: SXX stress under X loading

1/2:STRESS
Time: 1.000000000
Entity:SY

max: -4.70e+001
min: -1.22e+002

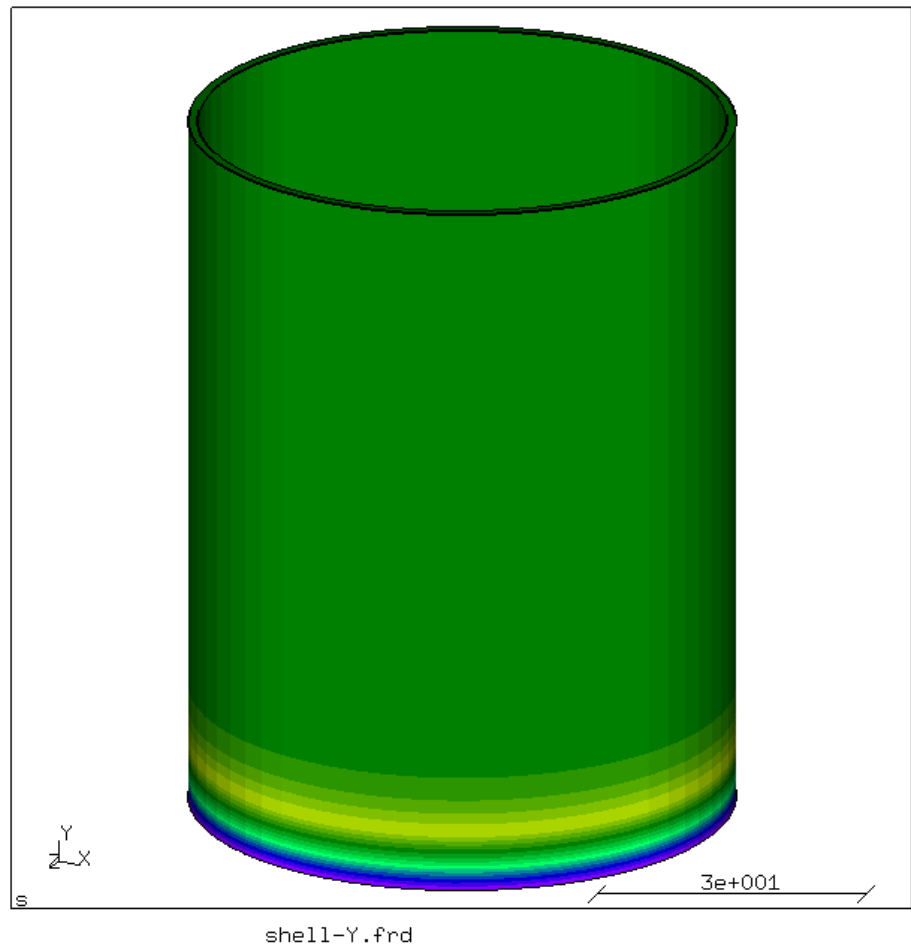
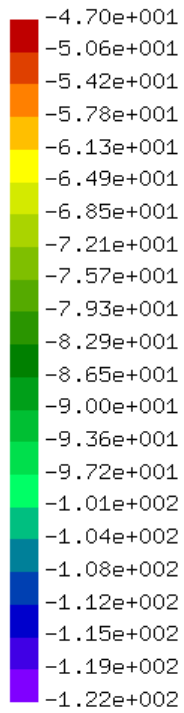


Figure 13: *SY* stress loading *Y* loading

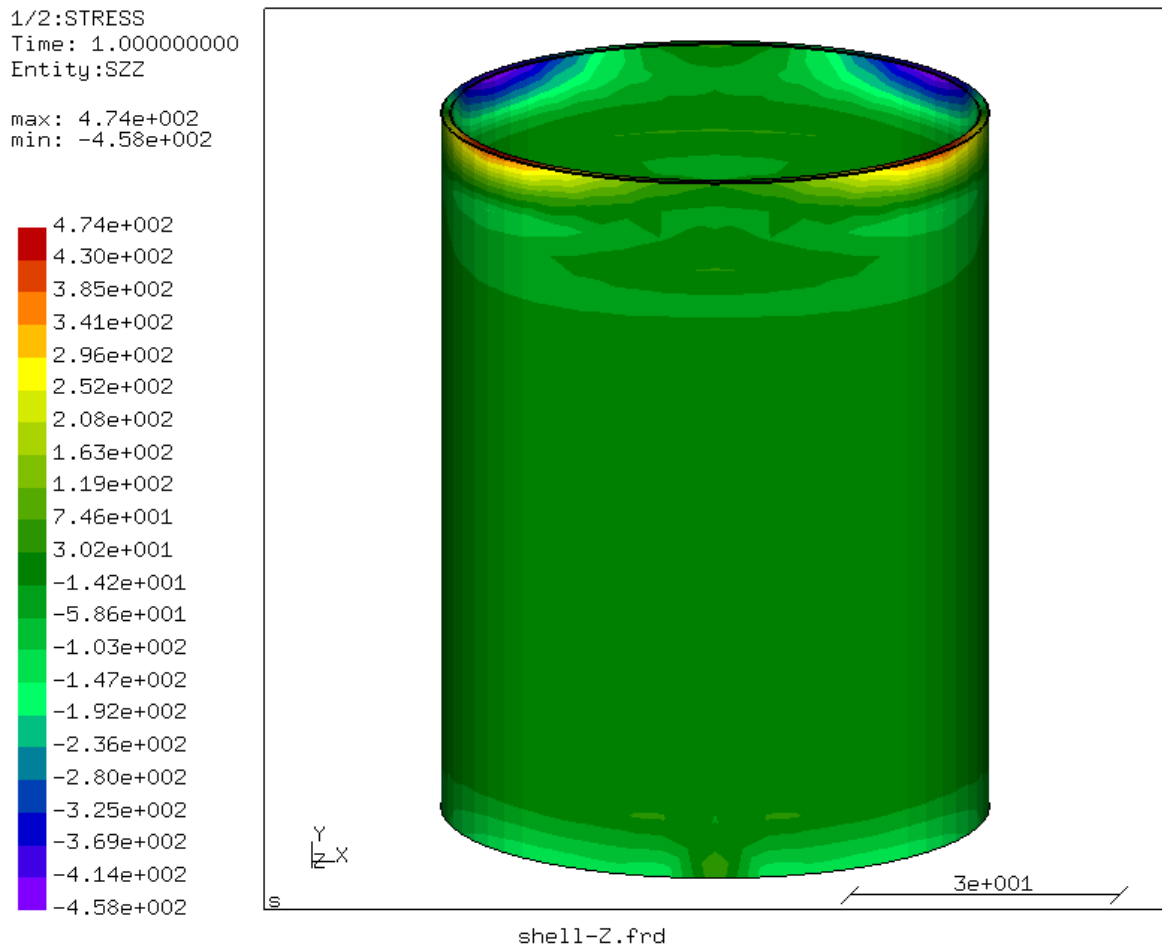


Figure 14: *SZZ stress under Z loading*

4.5. Transition from Shell Model to Solid Model

After completing the shell analyses, the model was converted into a solid cylindrical pipe model. This transition was necessary because the solid model represents the pipe wall thickness directly in the geometry, whereas the shell model represents thickness through a section property.

The solid model was created and meshed in CGX. Quadratic reduced-integration solid elements were used for the solid model. In this model, only translational degrees of freedom were constrained because standard solid elements do not have rotational degrees of freedom at their nodes.

The fixed end of the pipe was defined using the Nfix node set, and the loading end was defined using the Nload node set and Sload surface set.

The material properties used for the solid model were the same as the shell model:

- Elastic modulus: 210000 MPa
- Poisson's ratio: 0.3
- Density: 7.85E-9

The solid section was defined using the *SOLID SECTION keyword.

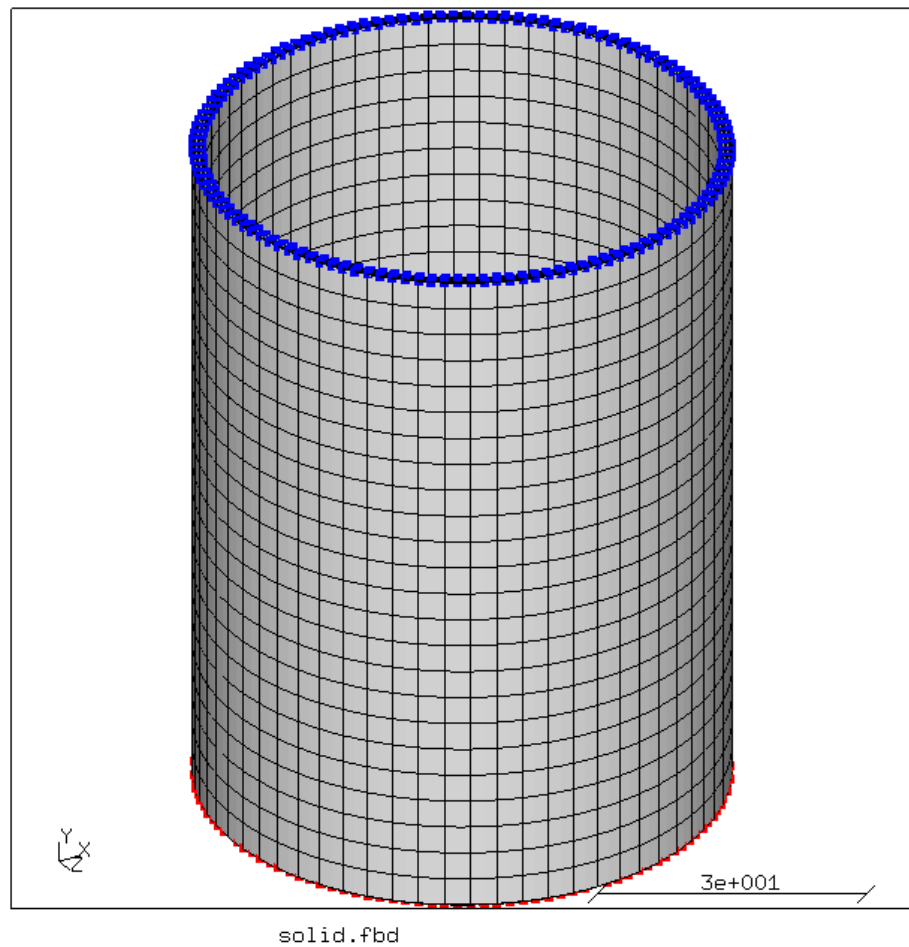


Figure 15: : Fixed region and loaded surface of the solid model

1/2:STRESS
Time: 1.000000000
Entity:SXX

max: 2.89e+002
min: -2.81e+002

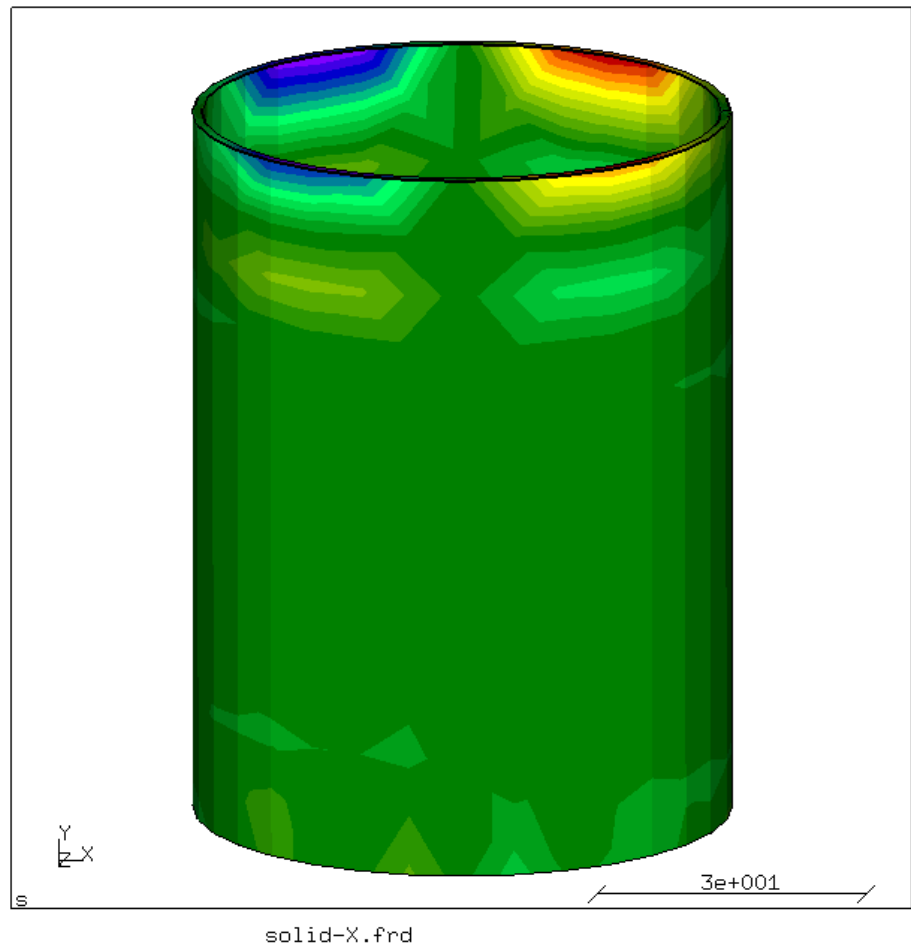
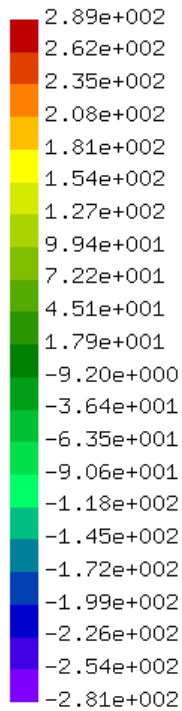


Figure 16: Normal stress component (S_{XX}) distribution of the solid cylindrical shell under X-direction loading.

1/2:STRESS
Time: 1.000000000
Entity:SY

max: 1.24e+002
min: -1.91e+000

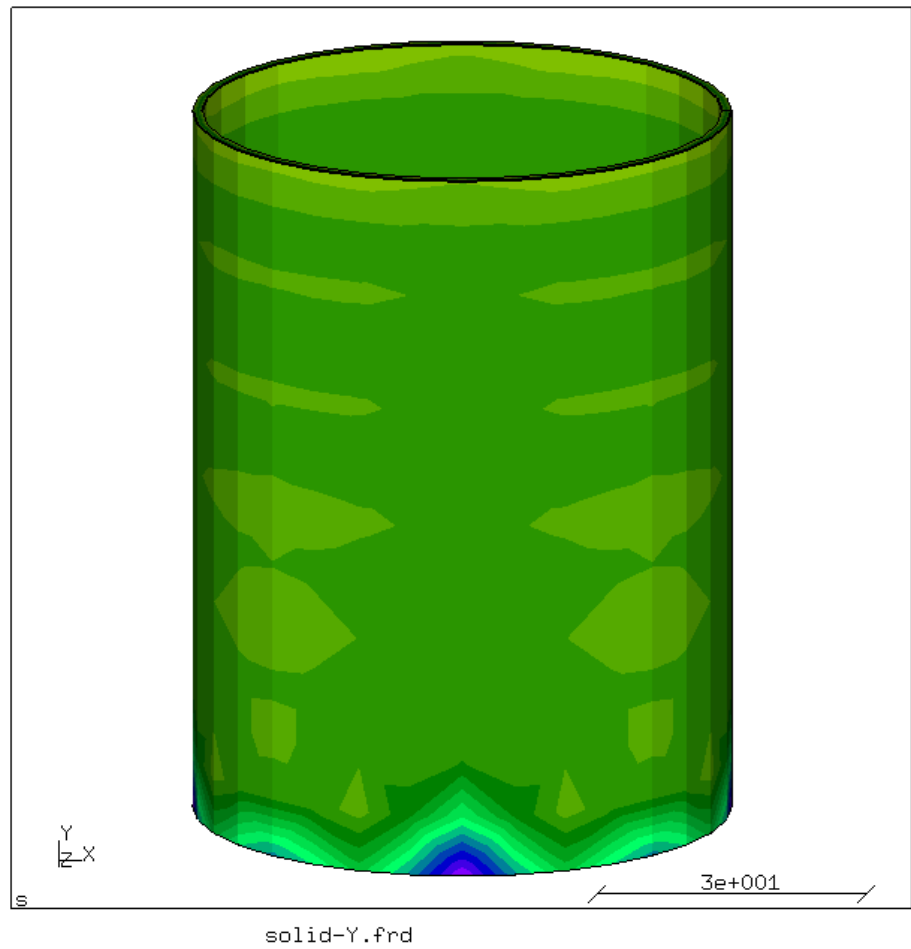
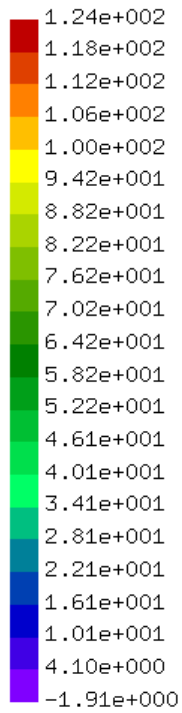


Figure 17: Normal stress component (S_{YY}) distribution of the solid cylindrical shell under Y-direction loading.

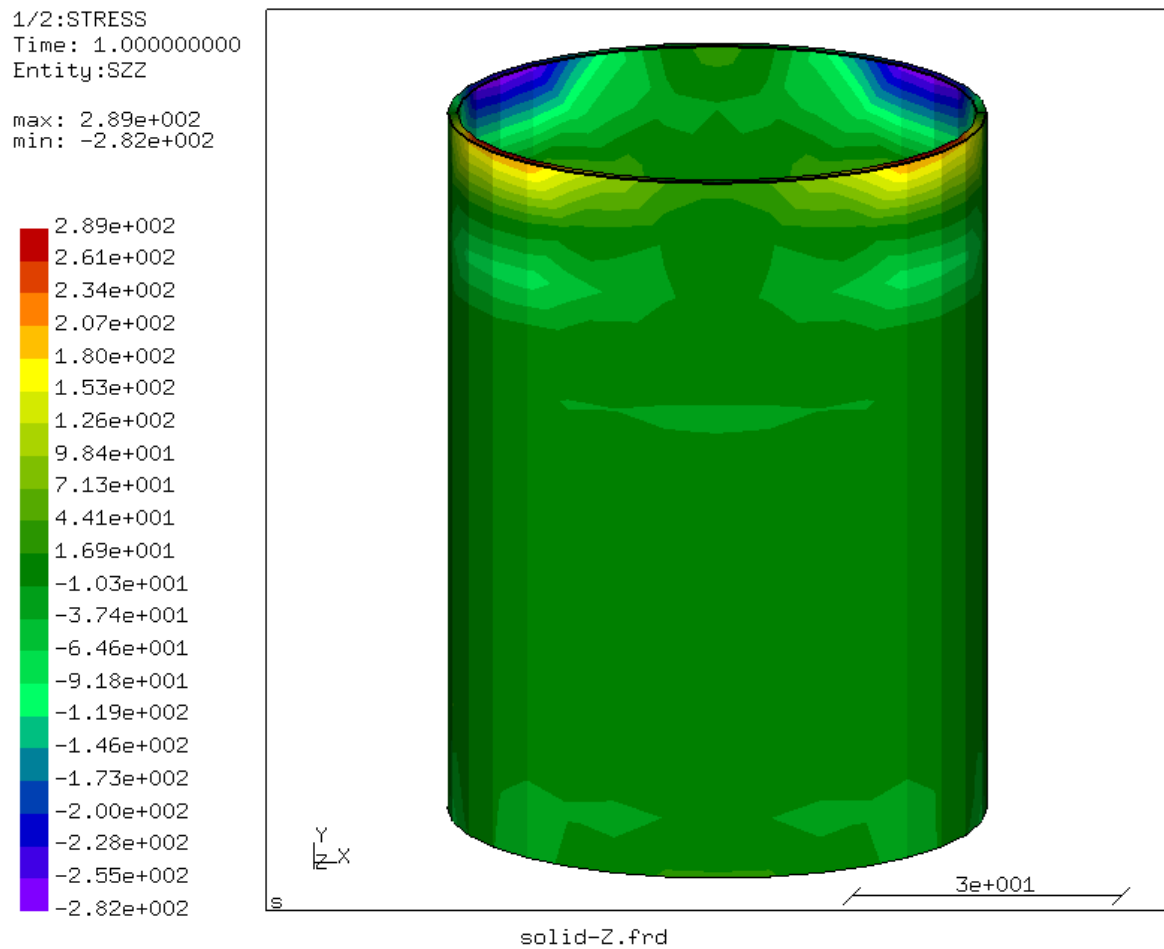


Figure 18: Normal stress component (S_{ZZ}) distribution of the solid cylindrical shell under Z-direction loading.

4.6. Remote Loading Method Using a Reference Node

After the solid model was prepared, a remote loading method was introduced. Instead of applying the load directly to all nodes on the loading edge, a reference node was created and coupled to the loading surface. This provided a cleaner and more controlled way to transfer the external load to the structure.

A reference node was defined near the loading end of the pipe:

999999, 0, 80, 0

This reference node was coupled to the loading surface using a kinematic coupling constraint:

```
*COUPLING, REF NODE=999999, SURFACE=Sload, CONSTRAINT NAME=remote_load
*KINEMATIC
1,3
```

With this method, the load was applied to the reference node and transferred to the loading surface through the coupling constraint. This approach also made it easier to apply both load-controlled and displacement-controlled boundary conditions in a consistent way.

4.7. Analytical Stress Verification

Before constructing the load–displacement curve, a basic analytical stress check was performed to verify whether the finite element model produced reasonable results.

The average stress was estimated using:

$$\sigma = \frac{F}{A}$$

where F is the total applied load and A is the cross-sectional area of the pipe. The cross-sectional area was calculated approximately as:

$$A = 2\pi r t$$

For one of the verification cases, the analytical stress value was approximately 84.9, while the corresponding CalculiX result was approximately -84.85 for the relevant stress component. The close agreement between the analytical and finite element results confirmed that the load transfer and model setup were reasonable.

4.8. Selection of Measurement Node

In order to extract displacement data for the load–displacement curve, a measurement node was selected from the loading region. Node 142 was selected as the measurement node.

The node was defined as a separate node set:

```
*NSET, NSET=NALI
```

```
142,
```

The Y-direction displacement of this node was extracted from the .dat file and used in the load–displacement curve. Since the loading was applied in the Y direction, the U2 displacement component was used.

4.9. Displacement-Controlled Analysis

The first method used to obtain the load–displacement response was the displacement-controlled method. In this method, prescribed Y-direction displacement values were applied to the loading region, and the corresponding reaction forces were obtained from the output file.

The displacement was applied using a boundary condition in the Y direction:

```
*BOUNDARY
```

```
Nload, 2, 2, -U
```

The reaction force was extracted from the .dat file using the reaction force output for the loading node set. The total Y-direction reaction force was used as the load value.

This method was used because displacement-controlled analysis is generally more stable for nonlinear structural problems near a limit point. In nonlinear buckling problems, load-controlled analysis may fail near the maximum load because the structure can lose stability. By prescribing displacement, the response can be followed more effectively up to the limit region.

The displacement-controlled results showed an initially linear increase in load with displacement. After that, the response became nonlinear and approached a limit-load region around 671 kN.

4.10. Load-Controlled Analysis

After the displacement-controlled analysis, a load-controlled analysis was performed. In this method, the load values obtained from the displacement-controlled analysis were applied directly through the reference node. Then, the resulting displacement was extracted from node 142.

The load was applied in the Y direction using:

```
*CLOAD
999999, 2, -P
```

For each load level, the analysis was executed and the Y-direction displacement U2 of node 142 was extracted from the .dat file. The same load levels from the displacement-controlled analysis were used in order to make a direct comparison between the two methods.

The load-controlled analysis converged successfully up to 671520 N. The corresponding displacement at this point was approximately 1.447091. However, for load values above this level, such as 671525 N, 671530 N, and 671550 N, the solver failed with the error:

increment size smaller than minimum

This indicated that the structure reached a nonlinear limit point and that the solver could no longer continue the equilibrium path using the current load-controlled static setup.

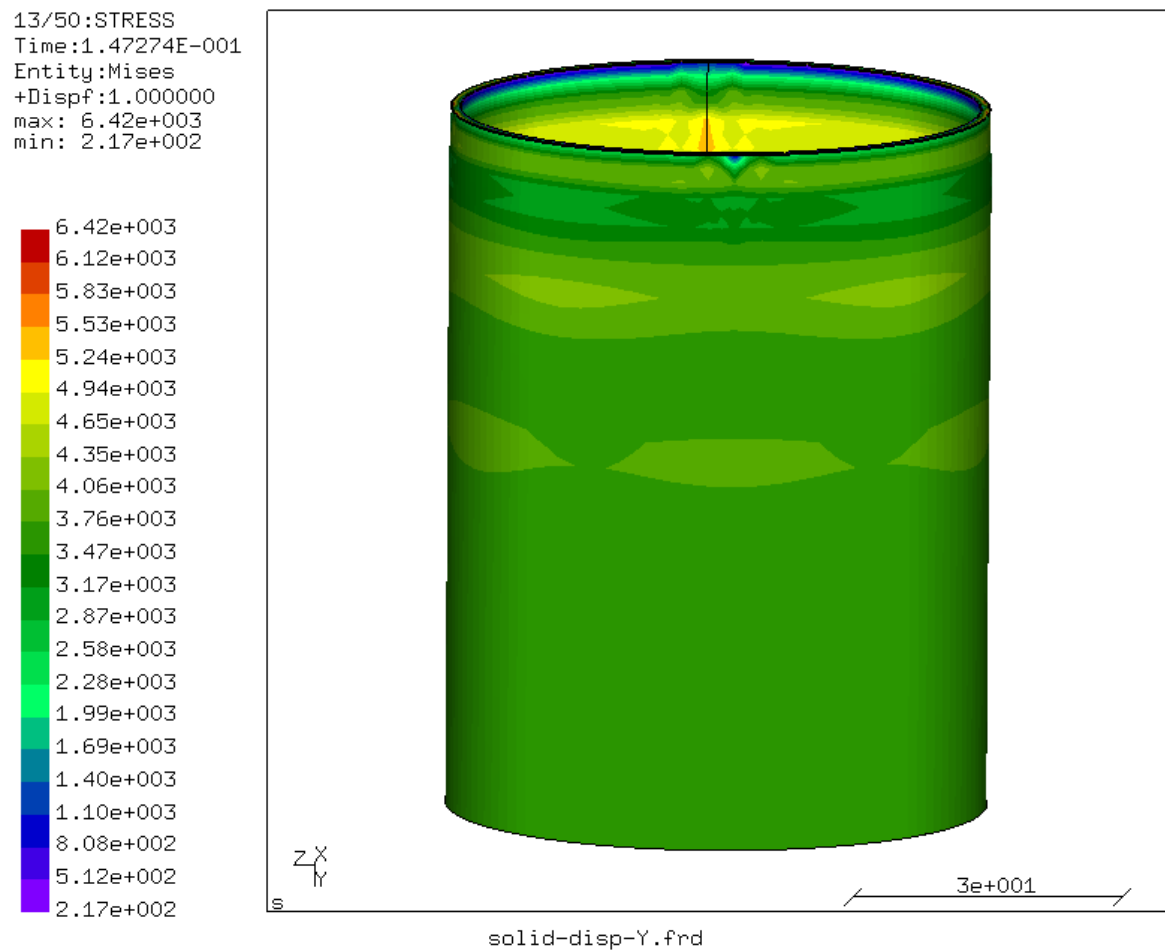


Figure 19:solid-disp-Y.frd

4.11. Comparison of Displacement-Controlled and Load-Controlled Results

The displacement-controlled and load-controlled results were plotted on the same load-displacement graph. The two curves showed very good agreement up to the limit region.

This confirmed that both methods predicted the same structural response in the converged range.

The final reliable load-controlled point was:

- Load: 671520 N
- Displacement: approximately 1.447091

The displacement-controlled method also showed convergence difficulties around the same displacement range. Additional displacement-controlled trials at larger displacement values, such as -1.450 and -1.470, did not complete the full step. The last converged states were again located around:

- Displacement: approximately 1.446–1.447
- Load: approximately 671.5–671.7 kN

Therefore, the approximate limit load of the structure was identified as:

$$P_{limit} \approx 671.52 \text{ kN}$$

with a corresponding displacement of approximately:

$$U_{limit} \approx 1.447$$

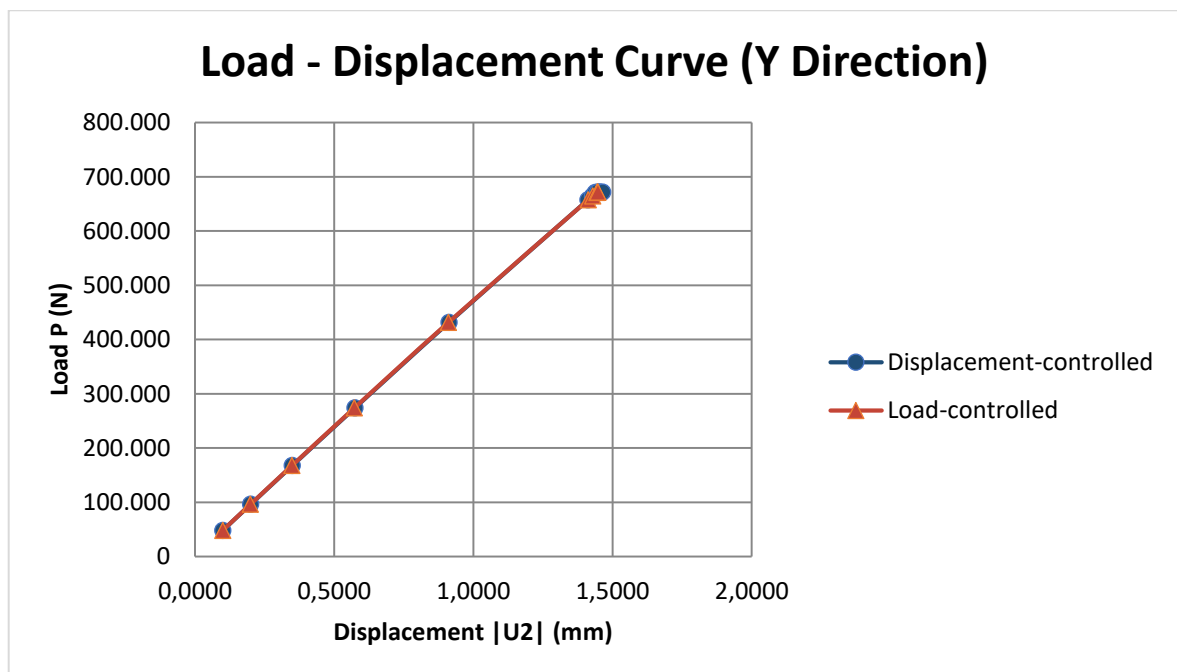


Figure 20: Final combined load–displacement curve

4.12. Limitation of the Current Static Analysis

Although both displacement-controlled and load-controlled methods successfully identified the limit region, the complete post-buckling branch of the curve could not be obtained using the current static analysis setup.

The solver stopped near the limit point with the error:

increment size smaller than minimum

This behavior indicates that the structure reached a nonlinear instability region. In this region, a standard static analysis may not be able to follow the descending or post-limit branch of the response.

Therefore, to obtain the complete post-buckling curve, further analysis methods may be required, such as:

- introducing an initial geometric imperfection,
- using a Riks or arc-length type post-buckling method,
- improving the displacement-control strategy through the reference node,
- using smaller and more controlled increments near the limit region.

In this project, the current methodology was sufficient to identify the approximate nonlinear limit load, but not sufficient to trace the full descending post-buckling response.

5. DISCUSSION

This section interprets the analytical and numerical results obtained for the thin-walled straight cylindrical shell and relates them to the buckling theory of Section 3 and to the findings reported in the literature survey of Section 2. Three characteristic axial load levels were obtained during the study: the analytical reference load based on an assumed critical stress, the linear eigenvalue (bifurcation) buckling load predicted by the finite element model, and the nonlinear limit load obtained from the displacement-controlled analysis. The aim of the discussion is to explain how these values relate to one another, why they differ, and what they reveal about the stability behaviour of the cylindrical shell.

5.1. Comparison of the Analytical, Eigenvalue, and Nonlinear Results

The analytical reference load adopted in the project was based on the assumed critical stress of 4000 MPa and the exact ring area of the cylindrical shell. The exact ring area was calculated as:

$$A = \pi[R^2 - (R - t)^2] \quad (1)$$

For the present geometry:

$$R = 30 \text{ mm}, \quad t = 1 \text{ mm}$$

Substitution of these values gives:

$$\begin{aligned} A &= \pi[30^2 - (30 - 1)^2] \\ A &= \pi(30^2 - 29^2) = 185.35 \text{ mm}^2 \end{aligned}$$

The analytical reference critical load was then obtained from:

$$P_{ref} = \sigma_{ref} A$$

(4)

where:

$$\sigma_{ref} = 4000 \text{ MPa}$$

Therefore:

$$P_{ref} = 4000 \times 185.35 = 741400 \text{ N}$$

$$P_{ref} \approx 741.4 \text{ kN}$$

The linear eigenvalue buckling analysis carried out in CalculiX returned a buckling factor of 81.67 for a reference load of 10 kN applied through the reference node. The finite element eigenvalue buckling load was therefore calculated as:

$$P_{cr,eig} = \lambda P_{ref,FE} \quad (7)$$

$$P_{cr,eig} = 81.67 \times 10 \text{ kN}$$

$$P_{cr,eig} \approx 816.7 \text{ kN}$$

Finally, the displacement-controlled nonlinear analysis produced a maximum, or limit, reaction force of 671.7 kN at an axial shortening of about 1.45 mm. These values are summarised in Table 1 and illustrated in Figure 7.1.

Method / source	P (kN)	Nature of the prediction
Linear eigenvalue buckling (LBA), $\lambda = 81.67$	816.7	Idealised perfect shell; upper-bound bifurcation load
Analytical reference, $\sigma_{ref} = 4000 \text{ MPa}$	741.4	Assumed reference critical stress multiplied by ring area
Nonlinear FE limit load (displacement-controlled)	671.7	Limit point of the perfect numerical model

Table 1: Comparison of the analytical, eigenvalue, and nonlinear axial load levels of the cylindrical shell.

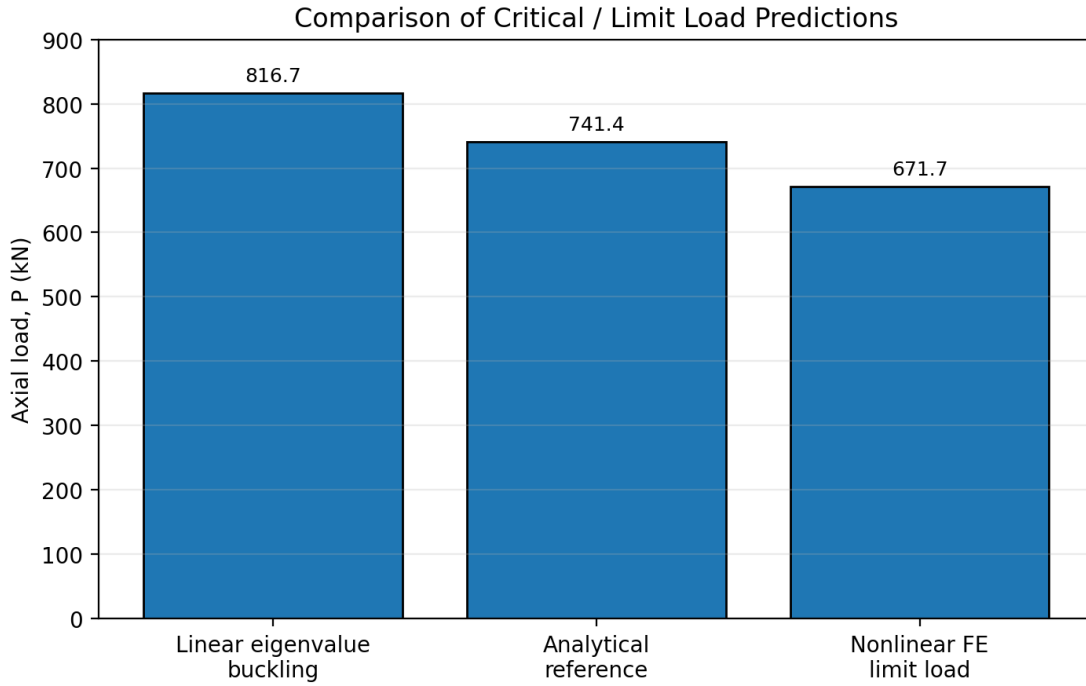


Figure 21: Comparison of the analytical reference, eigenvalue buckling, and nonlinear limit load predictions

The results show a clear and physically expected order. The ideal eigenvalue buckling load is the highest prediction because linear bifurcation analysis assumes a perfect geometry and small deformation behaviour. The analytical reference load is lower because it is based on the selected reference stress of 4000 MPa. The nonlinear limit load is the lowest value because the geometrically nonlinear analysis follows the actual deformation of the model and captures the reduction in stiffness near the instability region. The load hierarchy can be written as:

$$P_{FE,limit} < P_{analytical\ reference} < P_{eigenvalue}$$

$$671.7\text{ kN} < 741.4\text{ kN} < 816.7\text{ kN}$$

This ordering is consistent with the central message of the literature survey: linear bifurcation analysis assumes a perfect geometry and small deformations and therefore

predicts an idealised upper load, whereas a geometrically nonlinear analysis follows the actual deformation of the structure and predicts a lower limit load. The fact that the present results reproduce this hierarchy confirms that the analyses are behaving correctly rather than representing a coincidence of the chosen numbers.

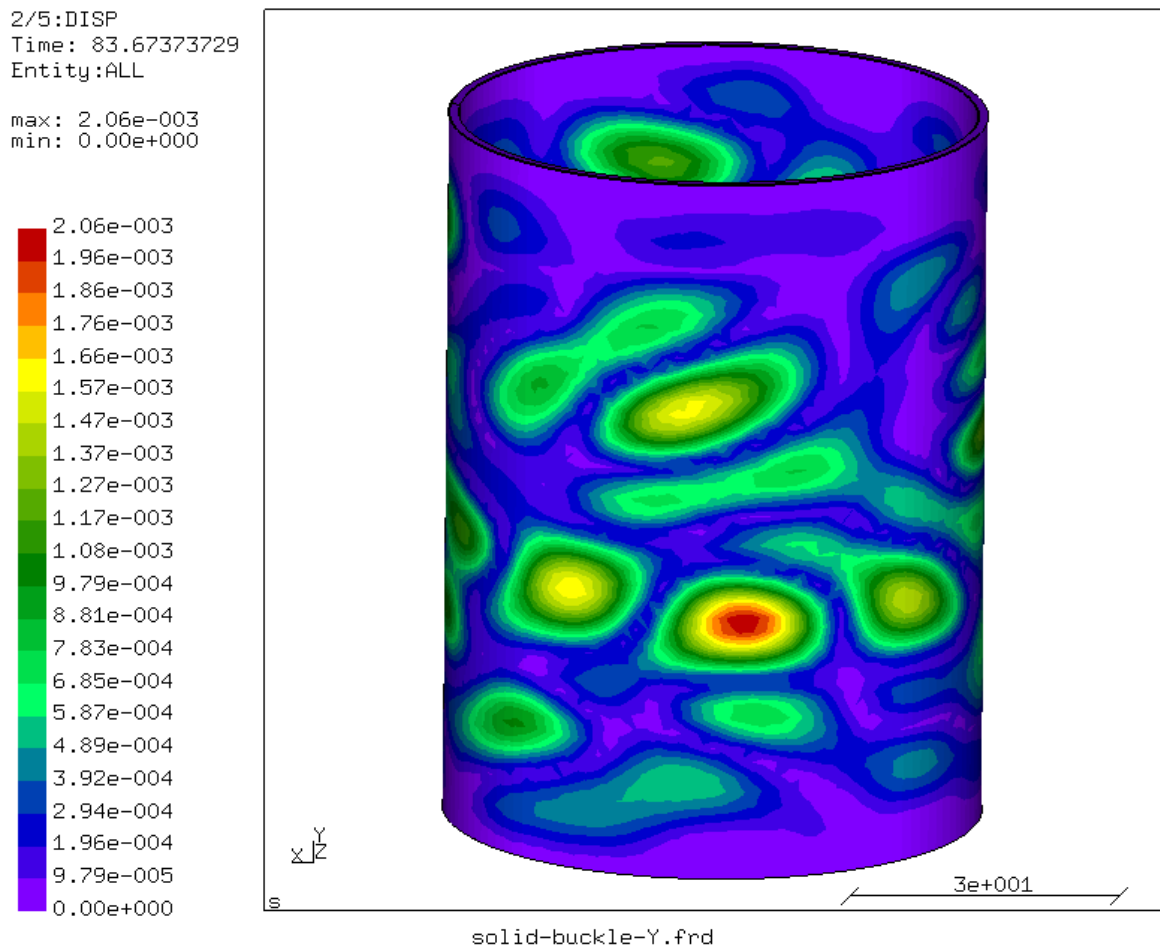


Figure 22: Buckling mode shape represented by the displacement contour from the linear eigenvalue buckling analysis of the solid cylindrical shell.

The displacement contour shown in Figure 17 represents the first buckling mode shape obtained from the linear eigenvalue buckling analysis. The contour does not indicate the actual physical displacement magnitude of the structure; instead, it shows the expected deformation pattern at the ideal bifurcation condition. Therefore, this result was used as a qualitative reference to understand the possible buckling shape of the cylindrical shell.

The eigenvalue analysis predicted an ideal buckling load of approximately 816.7 kN. However, this value is based on a perfect geometry, small deformation assumptions, and

linear elastic behavior. For this reason, it is expected to be higher than the nonlinear finite element result. In the nonlinear displacement-controlled analysis, the model reached the limit region at approximately 671.7 kN, which represents a more realistic limit load for the current numerical setup.

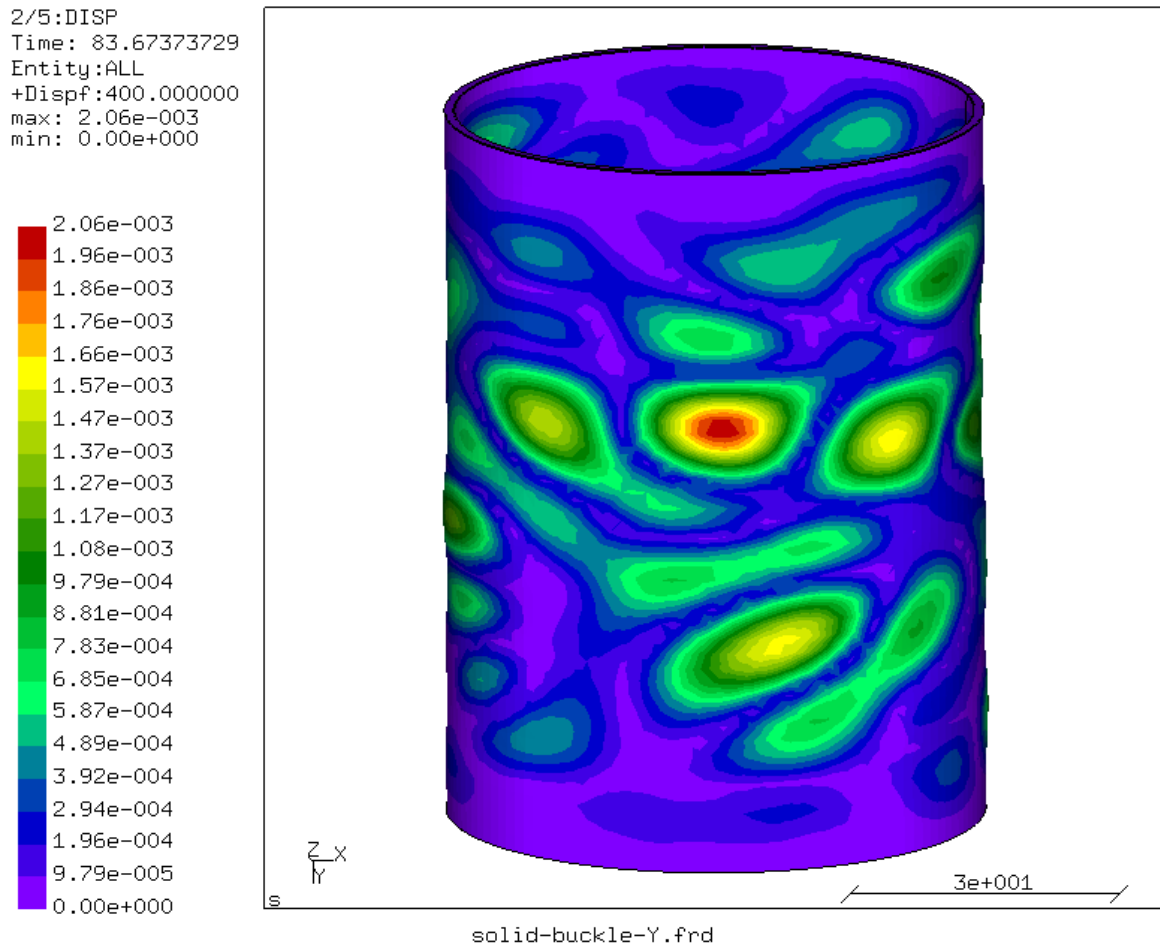


Figure 23: Scaled 400

The first eigenvalue buckling mode shape was also visualized using a deformation scale factor. In the original result, the displacement magnitude was very small and the deformation pattern was not clearly visible. Therefore, the deformation was scaled by using a scale factor of 400 in CGX. This scaling was applied only for visualization purposes and does not represent the actual physical displacement magnitude. The purpose of the scaled view is to

make the buckling mode shape and deformation tendency of the cylindrical shell easier to observe.

5.2. Interpretation of the Nonlinear Load-Displacement Response

The load-displacement curve extracted from the displacement-controlled analysis provides the clearest physical picture of the shell behaviour. The total reaction force was plotted against the prescribed end displacement as an XY scatter curve, as shown in Figure 7.2.

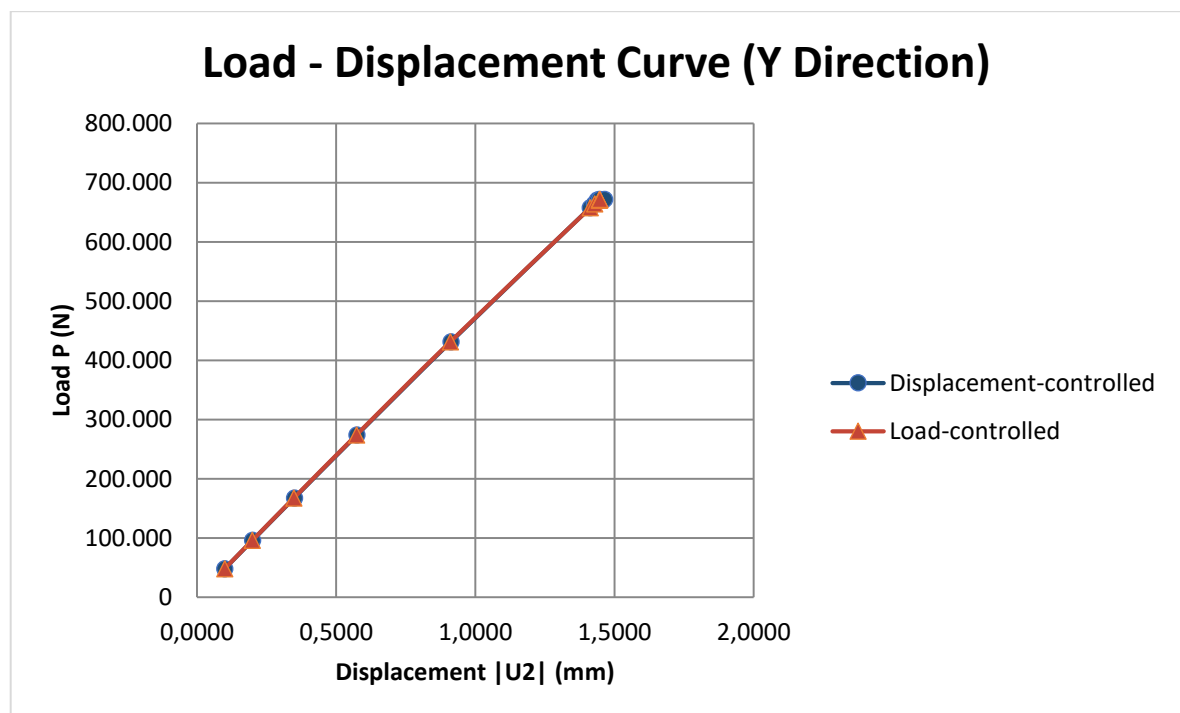


Figure 24: : Load-displacement response obtained from the displacement-controlled analysis

In the early part of the response the curve is essentially straight. The first computed point corresponds to a reaction force of 48.17 kN at an axial displacement of 0.10 mm. The initial axial stiffness was calculated as:

$$k_{initial} = \frac{P}{u} \quad (6)$$

$$k_{initial} = \frac{48.17 \text{ kN}}{0.10 \text{ mm}} = 481.7 \text{ kN/mm}$$

The elementary axial stiffness of the cylindrical shell is:

$$k_{axial} = \frac{EA}{L} \quad (1)$$

$$k_{axial} = \frac{210000 \times 185.35}{80} = 486.6 \text{ kN/mm}$$

The agreement between the numerical initial stiffness and the elementary axial stiffness confirms that, before instability sets in, the shell carries the axial load as a simple compression member and that the pre-buckling membrane response of the model is correct.

As the prescribed displacement increases, the slope of the curve gradually decreases. In terms of the nonlinear equilibrium framework of Section 3, this softening corresponds to a progressive reduction of the tangent stiffness matrix as the structure deforms. The response approaches a limit point at approximately 671.7 kN and 1.45 mm, beyond which the structure can no longer sustain an increasing load. This is the practical manifestation of the instability condition:

$$\det(\mathbf{K}_T) = 0$$

At the limit point the tangent stiffness becomes singular and the cylindrical shell loses its stable equilibrium configuration. The average axial stress at the nonlinear limit load was calculated as:

$$\sigma_{limit} = \frac{P_{limit}}{A}$$

$$\sigma_{limit} = \frac{671700}{185.35} = 3624 \text{ MPa}$$

The limit load obtained here is therefore the most representative estimate of the actual load-carrying capacity of the modelled perfect shell.

5.3. Force Control versus Displacement Control

An important methodological outcome of the project concerns the choice of solution control near the critical region. In the force-controlled remote-loading analysis, the compressive load was applied through the reference node and increased towards the

analytical reference value. However, the solver experienced convergence difficulties and stopped before the critical load was reached. This behaviour is not a sign that the model is wrong; it is the expected numerical consequence of approaching a limit point under load control.

Close to the limit point, the tangent stiffness tends to zero:

$$\mathbf{K}_T \rightarrow 0$$

Therefore, a small additional load increment may require a very large displacement increment:

$$\Delta \mathbf{u} = \mathbf{K}_T^{-1} \Delta \mathbf{F}$$

As a result, the load-controlled Newton-Raphson iteration can no longer find an equilibrium state on the rising branch. Switching to displacement control resolved this difficulty. In displacement control, the displacement is prescribed and the corresponding reaction force is calculated:

$$u_2 = u_{prescribed}, \quad R_F = R(u_{prescribed})$$

For the present displacement-controlled analysis, the imposed displacement was related to the step time by:

$$u_2 = -10 \times t_{step}$$

This allowed the analysis to advance through the stiffening response and reach the limit point, because displacement is a monotonic and controllable quantity even where the load is not. The concentration of very small, increasingly difficult increments near the final step is the numerical signature of the limit point. The contrast between the two control strategies demonstrates why instability problems must be solved with displacement control, or with an arc-length method, rather than with simple load control.

5.4. Effective Reduction and Comparison with the Literature

The nonlinear limit load is below the linear eigenvalue prediction. The effective reduction ratio was evaluated as:

$$\eta = \frac{P_{FE,limit}}{P_{eigenvalue}}$$

$$\eta = \frac{671.7}{816.7} = 0.82$$

It is important to interpret this ratio carefully. The finite element model used here is geometrically perfect: no initial imperfection was deliberately introduced into the mesh. The 18 percent reduction therefore arises solely from geometric nonlinearity and the prebuckling deformation of the perfect shell, and not from imperfection sensitivity. In the terminology of Section 3.3, the value 0.82 is closer to the ratio:

$$\frac{P_{nonlinear}}{P_{eigenvalue}}$$

rather than to a true knockdown factor:

$$\frac{P_{imperfect}}{P_{perfect}}$$

This distinction is exactly the point emphasised by the literature survey. NASA SP-8007 and the imperfection-sensitivity studies cited in Section 2 show that real cylindrical shells buckle well below the perfect-shell prediction once geometric imperfections, load eccentricity, and non-ideal boundary conditions are taken into account. Because the present analysis omits such imperfections, the limit load of 671.7 kN — and the associated ratio of 0.82 — should itself be regarded as an upper bound on the realistic capacity. A physical cylinder of the same geometry, manufactured with inevitable imperfections, would be expected to buckle at a still lower load. Viewed in this way, the present results are consistent with the knockdown-factor philosophy that motivates the project, and they provide the perfect-shell baseline to which an imperfection-sensitivity study could later be compared.

5.5. Limitations of the Present Study

Several idealisations limit the scope of the conclusions that can be drawn, and stating them explicitly clarifies what the results do and do not represent. First, the model is geometrically perfect, so imperfection sensitivity — identified in the literature as the dominant factor governing real cylindrical-shell buckling — has not yet been quantified. Second, the material was modelled as linear elastic with no plasticity; the critical stresses associated with the analytical and numerical buckling loads are much higher than the yield

and ultimate strength of structural steel. A real component of this geometry would therefore yield and collapse plastically before reaching these elastic buckling loads. The study should be understood as an idealised elastic-stability investigation rather than a prediction of physical failure load.

Third, the descending post-buckling branch of the response was not traced. CalculiX performs its nonlinear static solution with the Newton-Raphson method, which advances up to the limit point but cannot follow an equilibrium path on which the load decreases while the displacement continues to grow. Capturing that snap-back behaviour requires the arc-length or Riks method. For this reason, the post-buckling response in the present work is characterised up to the limit point, and the full descending branch remains a matter for future work. Finally, a single mesh density was used and a formal mesh-convergence study was not reported, and the loaded edge was represented either by reference-node coupling or by a uniform prescribed displacement, both of which are idealised boundary representations. None of these limitations undermines the main results, but each indicates a clear direction in which the analysis could be extended and made more realistic.

6. CONCLUSION

This project investigated the nonlinear buckling and post-buckling behaviour of a thin-walled straight cylindrical shell under axial compression using the open-source finite element package CalculiX together with its pre- and post-processor CGX. The aim, set out in the introduction, was to study the loss of stability of a thin cylindrical shell by combining analytical reference calculations with finite element analysis, and to understand how the predicted critical load depends on the assumptions of the analysis. That aim has been met.

The work followed a deliberately progressive workflow. The CalculiX and CGX file structure and command sequence were first learned on a preliminary elbow model, after which a straight cylinder was examined as a shell model with separate loading trials in the X, Y, and Z directions to verify the degree-of-freedom convention and the reading of displacement and stress output. The geometry was then rebuilt as a solid model, with $R =$

30 mm, $t = 1$ mm, and $L = 80$ mm, meshed with C3D20R twenty-node reduced-integration brick elements (4525 nodes and 1536 elements), so that the wall thickness was represented physically. The compressive load was transferred to the loaded end through a reference node coupled to the end surface for the force-controlled and eigenvalue analyses. On this model, an analytical reference calculation, a linear eigenvalue buckling analysis, a force-controlled nonlinear analysis, and a displacement-controlled nonlinear analysis were carried out, and the final load-displacement curve was extracted from the reaction-force output of the displacement-controlled run.

The principal quantitative results are coherent and mutually supporting:

$$P_{\text{eigenvalue}} \approx 816.7 \text{ kN}$$

$$P_{\text{analytical reference}} \approx 741.4 \text{ kN}$$

$$P_{FE, \text{limit}} \approx 671.7 \text{ kN}$$

The eigenvalue result gives the highest ideal load level, the analytical reference provides the project target based on the assumed critical stress, and the nonlinear limit load gives the lowest and most representative value for the geometrically nonlinear response of the perfect model. In addition, the initial slope of the nonlinear curve matches the elementary axial stiffness to within one percent, confirming the pre-buckling response. The loads fall in the physically expected order, with the nonlinear limit load below the idealised eigenvalue prediction.

Two engineering conclusions emerge from these results. First, near the critical region the choice of solution control is decisive: force control fails to converge as the tangent stiffness vanishes at the limit point, whereas displacement control advances through the instability and recovers the limit load and the load-displacement path up to that point. Second, even for a geometrically perfect shell, the nonlinear limit load is about eighteen percent below the linear eigenvalue prediction, and because real shells additionally suffer from geometric imperfections, the present perfect-shell value should be regarded as an upper bound on the realistic capacity. This result places the study firmly within the knockdown-factor framework of NASA SP-8007 and the imperfection-sensitivity literature reviewed in this report.

The study also has clear limitations, which define the natural next steps. The most valuable extension would be a geometrically nonlinear analysis with imperfections, in

which the first eigenvalue mode shape obtained from the linear buckling analysis is introduced as an initial geometric imperfection and its effect on the limit load is quantified; this would convert the present perfect-shell baseline into a realistic imperfection-sensitivity assessment. Beyond this, the descending post-buckling branch could be traced using an arc-length solution scheme, material plasticity could be added to obtain a physically meaningful collapse load, a mesh-convergence study could be performed to confirm numerical accuracy, and the numerical predictions could ultimately be compared against experimental or published buckling data. Taken together, the analyses completed in this project establish a verified finite element model and a consistent set of analytical and numerical buckling loads, providing a reliable foundation for these more advanced investigations and supporting the simulation-based, resource-conscious approach to structural design highlighted in the sustainability section of this thesis.

7. UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS

Sustainability is an important consideration in modern engineering design, especially in structures where safety, material efficiency, and long-term reliability are critical. In thin-walled cylindrical shell structures, buckling failure may occur suddenly and may lead to serious structural damage. Therefore, understanding the buckling behavior of cylindrical shells before manufacturing or physical testing is important for safer and more efficient design.

This study is mainly related to **SDG 9: Industry, Innovation and Infrastructure**. Cylindrical shell structures are used in many engineering applications such as pipelines, pressure vessels, tanks, aerospace structures, marine structures, and industrial systems. A better understanding of their buckling behavior contributes to the design of safer, more reliable, and more resilient engineering structures. In this sense, the project supports innovation in structural analysis and contributes to improved infrastructure design.

The project is also connected to **SDG 12: Responsible Consumption and Production**. By using finite element analysis, the number of physical trial-and-error experiments can be reduced. This helps decrease unnecessary material consumption and supports more efficient use of engineering resources. Instead of producing multiple physical prototypes, different loading cases and structural responses can be evaluated numerically.

In addition, the study indirectly supports **SDG 13: Climate Action**. Optimized structural design can reduce excessive material usage and support lightweight engineering solutions. Lower material consumption and fewer physical tests may reduce energy demand and environmental impact during the design and manufacturing stages. Although this project does not directly measure carbon emissions, it contributes to a simulation-based design approach that can support more sustainable engineering practices.

Therefore, this project contributes to sustainability by promoting numerical analysis, safer structural design, material efficiency, and reduced dependence on physical testing. The finite element investigation of cylindrical shell buckling provides a foundation for more reliable and resource-conscious engineering design.



Figure 25: United Nations Sustainable Development Goals.

8. DECLARATION OF ARTIFICIAL INTELLIGENCE USE

Advanced language models were used by us, Onur Sezen and Ali Kenan Kırmızıgül, exclusively for language refinement, formatting support, grammar correction, and improving the overall academic clarity of this thesis. These tools were not used to generate original numerical results, perform simulations, or replace our engineering interpretation. All final content, analyses, figures, tables, and conclusions were reviewed and approved by us.

9. REFERENCES

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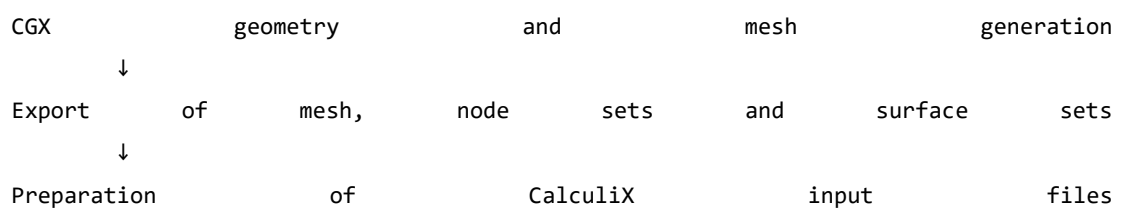
10. APPENDICES

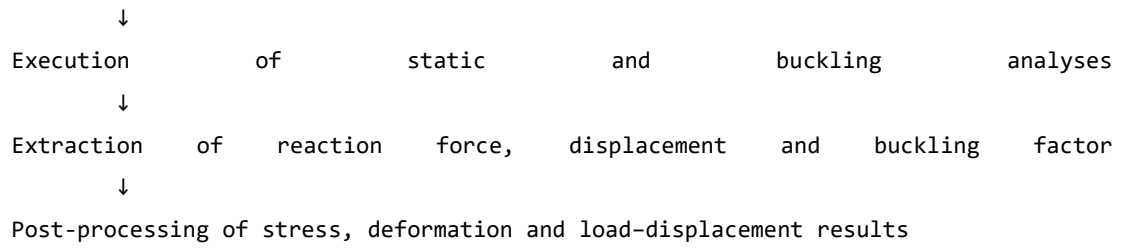
10.1. APPENDIX A: CalculiX and CGX File Structure Used in the Project

This appendix summarizes the main file types used during the finite element modelling and nonlinear buckling analysis of the thin-walled cylindrical shell. CalculiX CrunchiX was used as the solver, CalculiX GraphiX was used for pre-processing and post-processing, and Notepad++ was used for inspecting and editing the input and output files.

File type	Example file	Purpose in the project
.fbd	solid.fbd	CGX geometry and mesh generation file
.inp	solid-disp-Y.inp, solid-buckle-Y.inp, solid-remote-Y.inp	CalculiX input file defining material, boundary conditions, loading, output requests and analysis type
.msh	all.msh	Mesh file including node and element definitions
.nam	fix.nam, load.nam	Node set files used for the fixed boundary and loading region
.sur	load.sur	Surface definition used for reference node coupling
.dat	solid-disp-Y.dat, solid-buckle-Y.dat	Numerical output file including reaction forces, displacements and buckling factor results
.sta	solid-disp-Y.sta	Step and increment status file used to evaluate convergence behavior
.frd	solid-disp-Y.frd, solid-buckle-Y.frd	Result file opened in CGX for displacement, stress and mode shape visualization

The general workflow was:





10.2. APPENDIX B: Solid Cylindrical Shell Model Definition

The final numerical model was a solid thin-walled cylindrical shell. In this model, the wall thickness was represented physically by solid elements instead of being defined only as a shell section property.

Parameter	Symbol	Value
Outer radius	R	30 mm
Wall thickness	t	1 mm
Length	L	80 mm
Young's modulus	E	210000 MPa
Poisson's ratio	ν	0.3
Element type	—	C3D20R
Number of nodes	—	4525
Number of elements	—	1536

The mesh and boundary definitions were included in the CalculiX input file as:

```
*include,input=all.msh
*include,input=fix.nam
*include,input=load.nam
*include,input=load.sur
```

For the solid model, only translational degrees of freedom were fixed at the bottom end:

```
*BOUNDARY
Nfix,1,3
```

The steel material definition was:

```
*MATERIAL, NAME=steel
*ELASTIC
210000,0.3
```

```
*DENSITY
7.85E-9
```

The solid section assignment was:

```
*SOLID SECTION, ELSET=Ea11, MATERIAL=steel
```

10.3. APPENDIX C: Reference Node and Remote Loading Definition

For the force-controlled and linear eigenvalue buckling analyses, the load was applied through a reference node instead of being applied directly to every node in the loading region. This approach provided a controlled transfer of the compressive load to the end surface of the cylindrical shell.

The reference node was defined as:

```
*NODE
999999,0,80,0
```

The loading surface was defined in the load.sur file as:

```
*SURFACE, NAME=Sload
```

The reference node was coupled to the loading surface using:

```
*COUPLING, REF NODE=999999, SURFACE=Sload, CONSTRAINT NAME=remote_load
*KINEMATIC
1,3
```

The load transfer can be represented schematically as:

$$P_{ref} \rightarrow \text{Reference node} \rightarrow \text{Coupling constraint} \rightarrow S_{load}$$

where S_{load} is the coupled loading surface at the upper end of the cylindrical shell.

10.4. APPENDIX D: Analytical Reference Calculations

The cross-sectional area of the cylindrical shell was calculated using the exact ring area:

$$A = \pi[R^2 - (R - t)^2]$$

For the present geometry:

$$R = 30 \text{ mm}, \quad t = 1 \text{ mm}$$

Substitution gives:

$$A = \pi[30^2 - (30 - 1)^2]$$

$$A = \pi(30^2 - 29^2)$$

$$A = 185.35 \text{ mm}^2$$

The average axial stress under an applied compressive load is:

$$\sigma = \frac{P}{A}$$

where P is the applied axial load and A is the cross-sectional area.

The analytical reference load used in the project was based on the assumed critical stress:

$$\sigma_{ref} = 4000 \text{ MPa}$$

The corresponding analytical reference critical load was calculated as:

$$P_{cr,ref} = \sigma_{ref}A$$

$$P_{cr,ref} = 4000 \times 185.35$$

$$P_{cr,ref} = 741400 \text{ N}$$

$$P_{cr,ref} \approx 741.4 \text{ kN}$$

This value was used as an analytical reference point for evaluating the numerical buckling results.

10.5. APPENDIX E: Linear Eigenvalue Buckling Analysis

The linear eigenvalue buckling analysis was performed to obtain the ideal buckling load factor and the corresponding first buckling mode shape. The mathematical form of the finite element buckling eigenvalue problem is:

$$(\mathbf{K} + \lambda \mathbf{K}_G) \boldsymbol{\phi} = \mathbf{0}$$

where $\mathbf{K}$ is the linear elastic stiffness matrix, $\mathbf{K}_G$ is the geometric stiffness matrix generated by the applied reference load, λ is the eigenvalue or load multiplier, and $\boldsymbol{\phi}$ is the corresponding buckling mode shape.

The critical buckling load obtained from the eigenvalue analysis is:

$$P_{cr,eigen} = \lambda P_{ref}$$

In the CalculiX buckling input file, the static step was replaced by a buckling step:

```
*STEP,                               NLGEOM,                               INC=10000
*BUCKLE
1
```

The reference load was applied through the reference node:

```
*CLOAD
999999,2,-10000
```

Therefore:

$$P_{ref} = 10000 \text{ N} = 10 \text{ kN}$$

The first buckling factor obtained from solid-buckle-Y.dat was:

$$\lambda_1 = 83.67374$$

Thus, the ideal eigenvalue buckling load was:

$$P_{cr,eigen} = \lambda_1 P_{ref}$$

$$P_{cr,eigen} = 83.67374 \times 10000$$

$$P_{cr,eigen} = 836737.4 \text{ N}$$

$$P_{cr,eigen} \approx 836.7 \text{ kN}$$

This value represents the ideal linear buckling load of the perfect finite element model. Since eigenvalue buckling does not include geometric imperfections, large deformation effects or nonlinear stiffness changes, it should be interpreted as an upper-bound buckling prediction.

10.6. APPENDIX F: Force-Controlled Remote Static Analysis

The remote force-controlled nonlinear static analysis was performed using the reference node. The purpose of this analysis was to observe the response of the solid cylindrical shell under an applied compressive force.

The nonlinear static step was defined as:

```
*STEP,                                NLGEOM,                                INC=10000
*STATIC
```

The compressive load was applied through the reference node in the Y direction:

```
*CLOAD
999999, 2, -P
```

where P is the applied compressive load. The load-controlled analysis was able to approach the nonlinear limit region; however, convergence difficulties occurred near the instability point. This is consistent with the behavior of buckling problems because the tangent stiffness decreases as the critical region is approached.

The force-controlled equilibrium condition can be written as:

$$\mathbf{R}(\mathbf{u}, P) = \mathbf{F}_{int}(\mathbf{u}) - \mathbf{F}_{ext}(P) = \mathbf{0}$$

Near a limit point, the tangent stiffness becomes very small:

$$\det(\mathbf{K}_T) \rightarrow 0$$

As a result, load-controlled analysis may fail to continue the equilibrium path after the limit region.

10.7. APPENDIX G: Displacement-Controlled Nonlinear Static Analysis

After the convergence limitation observed in the force-controlled analysis, displacement-controlled nonlinear static analysis was used. In this method, displacement was prescribed to the loading region and the corresponding reaction force was obtained from the .dat output file.

The prescribed displacement was applied to the loading node set in the Y direction:

```
*BOUNDARY
Nload,2,2,-10
```

This means that the maximum prescribed displacement was -10 mm at step time 1.0. Therefore, the displacement at each increment was calculated as:

$$U_2 = -10 \times t_s$$

where t_s is the CalculiX step time.

The total reaction force of the loading node set was printed using:

```
*NODE          PRINT,          NSET=Nload,          TOTALS=YES
RF
```

The force value used in the final graph was the total reaction force in the Y direction:

$$P = \text{abs}(RF_y)$$

The displacement value used in the graph was:

$$u = \text{abs}(U_2)$$

Therefore, the final load-displacement curve was constructed using:

$$(|U_2|, |RF_y|)$$

The maximum nonlinear limit load obtained from the displacement-controlled analysis was:

$$P_{limit} \approx 671.7 \text{ kN}$$

at an axial shortening of approximately:

$$U_{limit} \approx 1.45 \text{ mm}$$

10.8. APPENDIX H: Load-Displacement Data Used for the Final Curve

The final load-displacement curve was obtained from the displacement-controlled nonlinear analysis. The negative signs in the raw output were caused by the selected loading direction. Therefore, absolute values were used for plotting.

Displacement, abs(U2) (mm)	Load, abs(RFy) (N)
0.1000	48174
0.2000	96137
0.3500	167681
0.5750	274075
0.9125	431449
1.4125	657598
1.4281	664296
1.4438	670746
1.4520	671295
1.4560	671425
1.4460	671550
1.4465	671706

The data were plotted using an XY scatter graph. The displacement values were placed on the horizontal axis and the reaction force values were placed on the vertical axis.

10.9. APPENDIX I: Summary of Main Numerical Results

The main analytical and numerical values obtained in this project are summarized below.

Result	Symbol	Value
Analytical reference critical load	$P_{cr,ref}$	741.4 kN
Linear eigenvalue buckling factor	λ_1	83.67374
Reference load in eigenvalue analysis	P_{ref}	10 kN

Result	Symbol	Value
Linear eigenvalue buckling load	$P_{cr,eigen}$	836.7 kN
Nonlinear limit load	P_{limit}	671.7 kN
Displacement at nonlinear limit load	U_{limit}	approximately 1.45 mm

The analytical reference value was used to estimate the expected critical region. The eigenvalue buckling analysis provided the ideal linear buckling load and first mode shape. The displacement-controlled nonlinear analysis provided the most representative limit load for the perfect finite element model.

